Double Advanced

Daily questions

Ten weeks of practice, reasoning and challenge

One day at a time

Most days have 20 questions. Wednesday has 25. Every day includes two multi-step challenges, labeled C 1 and C 2. The two challenges are included in the total.

Use separate paper

Write the day number and question label on your paper. Keep the packet open beside you. Draw number lines, grids and models when a question asks for them.

During weekend lessons

R is recall; P is a lesson problem; F is your turn; D is discussion; M is mixed review; X is extra practice; S is your summary; E is the exit ticket. Match the same label on the slide.

Pace the work

Allow about 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Work in two sittings when helpful. Weekend lessons allow 75-90 minutes for attempts, models, comparison and discussion.

Checks and references

Assessment pages are clearly labeled alternatives. Your tutor chooses them instead of the routine practice, or in a separate sitting. Sunday reference pages are for looking up an example; they are not another assignment.

Find a day

Print only the pages you need. Use PDF bookmarks to jump to a week or day.

Week	Sat	Sun	Mon	Tue	Wed	Thu	Fri
1	Day 001 p. 3-5	Day 002 p. 6-7	Day 003 p. 8	Day 004 p. 9	Day 005 p. 10	Day 006 p. 11	Day 007 p. 12
2	Day 008 p. 13	Day 009 p. 14-15	Day 010 p. 16	Day 011 p. 17	Day 012 p. 18	Day 013 p. 19	Day 014 p. 20
3	Day 015 p. 21	Day 016 p. 22-24	Day 017 p. 25	Day 018 p. 26	Day 019 p. 27	Day 020 p. 28	Day 021 p. 29
4	Day 022 p. 30	Day 023 p. 31-32	Day 024 p. 33	Day 025 p. 34	Day 026 p. 35	Day 027 p. 36	Day 028 p. 37
5	Day 029 p. 38	Day 030 p. 39-40	Day 031 p. 41	Day 032 p. 42	Day 033 p. 43	Day 034 p. 44	Day 035 p. 45
6	Day 036 p. 46	Day 037 p. 47-48	Day 038 p. 49	Day 039 p. 50	Day 040 p. 51	Day 041 p. 52	Day 042 p. 53
7	Day 043 p. 54	Day 044 p. 55-56	Day 045 p. 57	Day 046 p. 58	Day 047 p. 59	Day 048 p. 60	Day 049 p. 61
8	Day 050 p. 62	Day 051 p. 63-64	Day 052 p. 65	Day 053 p. 66	Day 054 p. 67	Day 055 p. 68	Day 056 p. 69
9	Day 057 p. 70	Day 058 p. 71-73	Day 059 p. 74	Day 060 p. 75	Day 061 p. 76	Day 062 p. 77	Day 063 p. 78
10	Day 064 p. 79	Day 065 p. 80-82	Day 066 p. 83	Day 067 p. 84	Day 068 p. 85	Day 069 p. 86	Day 070 p. 87

Fractions on one number line

20 questions, including 2 challenges. Write your work and answers on separate paper.

P1. Compare $\frac{3}{7}$ and $\frac{5}{7}$. Write <, >, or = .

P2. Compare $\frac{4}{5}$ and $\frac{4}{9}$. Use strips of the same length.

P3. Is $\frac{5}{12}$ less than or greater than $\frac{1}{2}$?

P4. Compare $\frac{3}{5}$ and $\frac{7}{12}$. Rename both using sixtieths.

P5. On a 0-to-2 line, place $\frac{5}{4}$, $1\frac{3}{4}$, and 0.5.

P6. Compare $\frac{7}{10}$ and 0.68.

P7. A line has ticks at 0, 2, and 4. Place $2\frac{1}{2}$.

P8. Three identical muffins are shared equally by four friends. How much muffin does each friend receive?

F. Complete: $\frac{5}{6}$ = __ twelfths. Compare $\frac{5}{6}$ and $\frac{3}{4}$.

D. Nora says $\frac{4}{9}$ is greater than $\frac{4}{5}$ because 9 is greater than 5. Explain and repair her claim.

X1. Compare $\frac{6}{14}$ and $\frac{8}{14}$. Explain using equal parts.

X2. Which is greater: $\frac{3}{5}$ or $\frac{3}{7}$? Explain the size of one part.

X3. Compare $\frac{7}{8}$ and $\frac{8}{9}$ using a common denominator.

X4. Compare $\frac{3}{4}$ and 0.74. Rename one number.

X5. On a 0-to-2 number line, place $\frac{11}{8}$. Name its mixed-number form.

X6. Name a fraction with denominator 8 strictly between $\frac{1}{2}$ and $\frac{3}{4}$.

S. To compare fractions, I first keep ___ the same. Then I can ___.

E. Order $\frac{3}{4}$, $\frac{7}{10}$, and 0.72 from least to greatest. Place them on a 0-to-1 line.

CHALLENGE / Explain your plan and check the result

C1. Ari finishes $\frac{3}{5}$ of a trail and Bea finishes $\frac{7}{12}$ of the same trail. Who is farther? Find the difference and show both positions on one line.

C2. Lina has $\frac{5}{8}$ liter and Omar has 0.6 liter. Each wants $\frac{3}{4}$ liter. Who needs less extra liquid? Find both gaps.

Optional oral fact snapshot

Use a short oral sample to spot facts needing practice. This is not extra daily homework.

1. 5 × 4	2. 3 × 6	3. 7 × 11
4. 7 × 8	5. 8 × 8	6. 4 × 3
7. 7 × 6	8. 5 × 7	9. 8 × 11
10. 2 × 11	11. 7 × 9	12. 6 × 9
13. 8 × 2	14. 12 × 10	15. 4 × 5
16. 7 × 4	17. 3 × 5	18. 2 × 4
19. 4 × 12	20. 6 × 10	21. 11 × 2
22. 2 × 5	23. 6 × 11	24. 6 × 3
25. 8 × 7	26. 6 × 12	27. 7 × 10
28. 8 × 6	29. 7 × 2	30. 12 × 7

Diagnostic: computation and reasoning

Tutor choice: use this check in place of the routine questions, or in a separate assessment sitting. Keep the two daily challenges for discussion.

1. 426×23

2. $1{,}368 \div 6$

3. $2{,}184 \div 24$

4. $4.65 + 13.8$

5. $8.1 - 3.47$

6. 0.7×0.4

7. $6.3 \div 0.9$

8. $\frac{3}{5} \times \frac{5}{6}$

9. $\frac{5}{6} + \frac{1}{4}$

10. How many $\frac{1}{4}$ s fit in 3?

11. $\frac{1}{5} \div 3$

12. $2\frac{1}{3} \times 3$

13. Compare $\frac{4}{9}$ and $\frac{1}{2}$.

14. Place $\frac{3}{4}$, $\frac{4}{3}$, $\frac{5}{8}$, and $1\frac{1}{2}$ on the line.

15. Order 0.6, 0.56, and 0.506 from least to greatest.

16. A bag has 3 red tiles for every 2 blue. There are 15 red tiles. How many blue tiles?

17. Find 25% of 64.

18. Evaluate $4n + 3$ when $n = 6$.

19. Morning temperature is $-4°C$. Noon is $2°C$. How much warmer is noon?

20. A rectangle is $\frac{3}{4}$ m long and $\frac{2}{3}$ m wide. Find its area.

21. A box is 4 cm by 3 cm by 5 cm. Find its volume.

22. Plot A(2, 4) and B(5, 4). How far apart are they?

23. Explain why $\frac{2}{3} \times 9$ is smaller than 9.

24. Persistence problem. A pattern starts 2, 5, 9, 14. Each jump grows by 1. Find the next two terms.

25. Persistence problem. Three boxes hold 18 tiles. Each box has two more than the box before it. Find the three counts.

Fraction operations with models

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. 7×8

R2. $96 \div 12$

R3. $\frac{1}{2} + \frac{1}{4}$

R4. $0.6 + 0.25$

P1. Five bags each hold $\frac{3}{4}$ cup of seeds. How many cups altogether?

P2. A rectangle is $\frac{3}{5}$ meter long and $\frac{2}{3}$ meter wide. Find its area using equal parts.

P3. Find $\frac{2}{3}$ of $1\frac{1}{2}$ cups.

P4. Without multiplying first, compare $\frac{3}{4} \times 8$ and $\frac{5}{4} \times 8$ with 8. Then compute.

P5. Three meters of ribbon are cut into $\frac{1}{4}$-meter pieces. How many pieces?

P6. A $\frac{1}{4}$-liter juice sample is shared equally among three cups. How much goes in each cup?

P7. Write one story for $2 \div \frac{1}{5}$ and one for $\frac{1}{5} \div 2$. Solve both.

F. A model has 4 columns and 3 rows. Shade $\frac{3}{4}$, then $\frac{2}{3}$ of it. Complete __ × __ = __.

D. Eli says $3 \div \frac{1}{4} = \frac{3}{4}$, because division makes numbers smaller. Show why this fails.

M1. 13×4

M2. $72 \div 9$

M3. $\frac{1}{3} + \frac{1}{6}$

S. When I divide, I first ask whether I need the number of ___ or the size of ___.

E. A board is 5 meters long. Each shelf needs $\frac{1}{2}$ meter. How many shelves can be cut?

CHALLENGE / Explain your plan and check the result

C1. Two batches each use $\frac{3}{4}$ liter of smoothie. All the smoothie is poured into $\frac{1}{4}$-liter cups. How many full cups? Explain with a model.

C2. A small rug is $1\frac{1}{2}$ meters by $\frac{2}{3}$ meter. A mat has area $\frac{4}{5}$ square meter. Which covers more floor, and by how much?

Week 1 reference

The whole stays the same when fractions are compared. Multiplication can take part of an amount. Division can count fractional groups or split a fraction into shares.

Fractions on one number line

A fraction names equal parts of one whole. The size of that whole stays fixed.

Example: Compare $\frac{2}{3}$ and $\frac{3}{4}$.

Use one whole for both fractions.

Split the whole into 12 equal parts.
$\frac{2}{3} = \frac{8}{12}$; $\frac{3}{4} = \frac{9}{12}$.

Eight twelfths is less than nine twelfths.

Result: $\mathbf{\frac{2}{3} < \frac{3}{4}}$

Fraction operations with models

Multiplication can take part of an amount. Division can count groups or find the size of each share.

Example: Find $\frac{3}{4}$ of $\frac{2}{3}$ cup.

Split a whole into 3 equal columns and 4 equal rows.

Select 2 columns, then 3 fourths of those columns.

The overlap is 6 of 12 pieces: $\frac{6}{12} = \frac{1}{2}$.

Result: $\mathbf{\frac{1}{2}}$ **cup**

Words to use

Numerator: The top number counts equal parts.

Denominator: The bottom number names how many equal parts make one whole.

Equivalent fractions: Different names for the same amount.

Quotient: The result of division.

Scaling: Multiplying an amount to change its size.

Fractions and equal parts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 3×3

2. 3×7

3. 3×4

4. 3×9

5. $24 \div 2$

6. 8×5

7. 4×4

8. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

9. Compare $\frac{2}{9}$ and $\frac{7}{9}$.

10. Compare $\frac{2}{4}$ and $\frac{2}{5}$.

11. Order $\frac{2}{5}$, 0.45, and $\frac{1}{2}$ from least to greatest.

12. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

13. Nora says $\frac{4}{9}$ is greater than $\frac{4}{5}$ because 9 is greater than 5. Explain and repair her claim.

14. Compare $\frac{9}{14}$ and $\frac{11}{14}$. Explain using equal parts.

15. Which is greater: $\frac{2}{6}$ or $\frac{2}{8}$? Explain the size of one part.

16. Compare $\frac{3}{7}$ and $\frac{5}{12}$ using a common denominator.

17. Compare $\frac{1}{4}$ and 0.24. Rename one number.

18. On a 0-to-2 number line, place $\frac{10}{6}$. Name its mixed-number form.

CHALLENGE / Explain your plan and check the result

C1. Jade walks $\frac{3}{8}$ mile, then $\frac{1}{4}$ mile. Sam walks $\frac{2}{3}$ mile. Who walks farther? Find the gap.

C2. A jar is $\frac{5}{6}$ full. Pour out $\frac{1}{4}$ of the jar's full capacity. Is the amount left above or below half? By how much?

Fractions and equal parts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. 3×5

6. 2×6

7. $60 \div 5$

8. Find both $4 \div \frac{1}{5}$ and $\frac{1}{5} \div 4$.

9. Find $3 \times \frac{2}{3}$.

10. $\frac{2}{3} \times \frac{2}{5}$

11. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

12. Share $\frac{1}{3}$ liter equally among 2 cups. How much per cup?

13. Eli says $3 \div \frac{1}{4} = \frac{3}{4}$, because division makes numbers smaller. Show why this fails.

14. 5 bags each hold $\frac{3}{6}$ kilogram. Find the total mass.

15. Find $\frac{5}{6}$ of $\frac{4}{8}$. Sketch equal parts.

16. Find $\frac{1}{5}$ of $1\ \frac{1}{4}$ liters.

17. A 7-meter cord is cut into $\frac{1}{8}$-meter pieces. How many pieces?

18. Share $\frac{1}{3}$ liter equally among 3 cups. How much per cup?

CHALLENGE / Explain your plan and check the result

C1. A baker has 3 meters of ribbon. Four packages each need $\frac{5}{8}$ meter. Is there enough ribbon? Find the amount left or missing.

C2. A $\frac{1}{3}$-liter sample is shared equally among 4 cups. Another sample gives each cup $\frac{1}{10}$ liter. Which share is larger, and by how much?

Fractions and equal parts

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 3.2×10

2. $300 \div 100$

3. Double 3.5

4. Half of 18

5. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

6. Write a story for $\frac{1}{6} \div 3$ and solve it.

7. 4×6

8. $\frac{1}{6} + \frac{1}{6}$

9. $300 \div 100$

10. 3×2

11. 6×6

12. A box is 6 cm by 6 cm by 3 cm. Find its volume.

13. A card shows a fraction. Its denominator is 8. It lies between $\frac{1}{2}$ and $\frac{3}{4}$. What fraction is on the card?

14. Compare $\frac{2}{11}$ and $\frac{4}{11}$. Explain using equal parts.

15. Which is greater: $\frac{2}{5}$ or $\frac{2}{7}$? Explain the size of one part.

16. Compare $\frac{5}{12}$ and $\frac{4}{9}$ using a common denominator.

17. Compare $\frac{1}{4}$ and 0.24. Rename one number.

18. On a 0-to-2 number line, place $\frac{10}{6}$. Name its mixed-number form.

19. Name a fraction with denominator 20 strictly between $\frac{1}{2}$ and $\frac{3}{4}$.

20. Complete $\frac{1}{6} = \frac{?}{30}$. Explain your multiplier.

21. Find $\frac{2}{5} + \frac{1}{3}$. Use equal-sized parts.

22. Find $\frac{5}{6} - \frac{1}{3}$. Check by addition.

23. Use a multiplication fact to find $70 \div 7$.

CHALLENGE / Explain your plan and check the result

C1. A card shows n twelfths. Its value is greater than $\frac{1}{2}$ and less than $\frac{3}{4}$. Find every whole-number value of n, then order those fractions.

C2. Three pitchers each hold $\frac{2}{3}$ liter. After $\frac{1}{2}$ liter is served, the rest fills $\frac{1}{4}$-liter cups. How many cups are filled?

Fractions and equal parts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 348 to the nearest hundred.

2. Round 3.72 to the nearest whole number.

3. Use 20×5 to estimate 19.8×5.1.

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

5. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

6. Write a story for $\frac{1}{6} \div 3$ and solve it.

7. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

8. A box is 5 cm by 2 cm by 3 cm. Find its volume.

9. 6×8

10. $48 \div 6$

11. 7×9

12. $63 \div 7$

13. Is $\frac{2}{3} \times 9$ smaller or larger than 9? Compute to check.

14. Eli says $3 \div \frac{1}{4} = \frac{3}{4}$, because division makes numbers smaller. Show why this fails.

15. 7 bags each hold $\frac{2}{5}$ kilogram. Find the total mass.

16. Find $\frac{4}{5}$ of $\frac{3}{6}$. Sketch equal parts.

17. Find $\frac{3}{5}$ of $1\frac{1}{3}$ liters.

18. A 7-meter cord is cut into $\frac{1}{4}$-meter pieces. How many pieces?

CHALLENGE / Explain your plan and check the result

C1. A garden is $2\frac{1}{2}$ meters long and $\frac{3}{5}$ meter wide. Half its area grows herbs. Find the garden area and herb area.

C2. Kai has $\frac{7}{8}$ meter of fabric and uses $\frac{1}{4}$ meter. Ren has 0.7 meter and uses 0.1 meter. Who has more left? Find the difference.

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

2. Write a story for $\frac{1}{6} \div 3$ and solve it.

3. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

4. Find $\frac{1}{2} + \frac{1}{3}$.

5. A box is 6 cm by 3 cm by 3 cm. Find its volume.

6. $72 \div 6$

7. $6.5 + 0.25$

8. $500 \div 100$

9. 5×4

10. Nora says $\frac{4}{9}$ is greater than $\frac{4}{5}$ because 9 is greater than 5. Explain and repair her claim.

11. Compare $\frac{8}{11}$ and $\frac{10}{11}$. Explain using equal parts.

12. Which is greater: $\frac{2}{4}$ or $\frac{2}{6}$? Explain the size of one part.

13. Compare $\frac{2}{3}$ and $\frac{7}{12}$ using a common denominator.

14. Compare $\frac{3}{8}$ and 0.365. Rename one number.

15. On a 0-to-2 number line, place $\frac{10}{6}$. Name its mixed-number form.

16. Name a fraction with denominator 20 strictly between $\frac{1}{2}$ and $\frac{3}{4}$.

17. Complete $\frac{6}{10} = \frac{?}{30}$. Explain your multiplier.

18. Find $\frac{1}{4} + \frac{1}{3}$. Use equal-sized parts.

CHALLENGE / Explain your plan and check the result

C1. A recipe needs $\frac{2}{3}$ cup of flour per batch. Make 3 batches from a $2\frac{1}{2}$-cup bag. How much flour remains? Is that more or less than one batch needs?

C2. Mira says $\frac{4}{5}$ of 10 must be smaller than $\frac{3}{4}$ of 12 because $\frac{4}{5}$ is greater than $\frac{3}{4}$. Check both products and explain what her comparison overlooks.

Equivalent ratios

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Share $\frac{1}{3}$ liter equally among 2 cups. How much per cup?

R2. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

R3. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

R4. Find $\frac{3}{4} \times \frac{4}{5}$.

P1. There are 4 red tiles and 3 blue tiles. Write red : blue. Write blue : red.

P2. A trail mix uses 4 raisins for every 3 nuts. How many nuts go with 12 raisins?

P3. A box has 20 red tiles and 15 blue tiles. Divide both counts by 5 to find an equivalent ratio.

P4. A recipe uses 6 cups of oats for every 4 cups of seeds. How many cups of seeds go with 9 cups of oats?

P5. Paint uses 2 cups yellow for every 5 cups blue. How much yellow goes with 15 cups blue?

P6. Each game team has 3 builders for every 2 testers. How many builders go with 8 testers?

F. A mix uses 5 cups oats for every 2 cups seeds. Ten cups oats makes __ batches and needs __ cups seeds.

D. Mia changes 4 red : 3 blue to 7 red : 6 blue by adding 3 to each count. Is the ratio unchanged?

X1. Red: blue is 2 : 7. Complete (2, 7), (4,__).

X2. A mix has 15 raisins and 35 nuts. Divide both counts by 5; write the new ratio.

X3. A recipe uses 4 cups oats for 10 cups seeds. How many cups seeds go with 6 cups oats?

X4. Is 8 : 10 equivalent to 3 : 5? Explain without using the difference.

S. To keep a ratio the same, I ___ both quantities by ___.

E. A bracelet pattern uses 7 green beads for every 4 white beads. How many green beads go with 12 white beads?

CHALLENGE / Explain your plan and check the result

C1. A mosaic uses 3 red tiles for every 5 blue tiles. You use 18 red tiles and have 32 blue tiles. How many blue tiles are needed? How many remain?

C2. A trail mix uses 4 cups oats for 6 cups seeds. You have 10 cups oats and 18 cups seeds. Use all the oats in the same ratio. How many cups seeds remain?

Ratio tables and comparisons

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

R2. Find $\frac{3}{4} - \frac{1}{3}$.

R3. Is $\frac{2}{3} \times 9$ smaller or larger than 9? Compute to check.

R4. Find $3 \times \frac{2}{3}$.

P1. Concentrate : water is 2 : 5. Complete pairs (2, 5), (4, __), and (6, 15).

P2. For that 2 : 5 mixture, how much water matches 10 cups concentrate? Solve using a table and a double number line.

P3. Mix A uses 3 cups syrup to 4 cups water. Mix B uses 5 cups syrup to 8 cups water. Which has more syrup for the same water?

P4. Yellow : blue paint is 2 : 3. There are 35 cups in all. How much of each color?

P5. Plot (1, 3), (2, 6), and (3, 9). Let x be bags and y be marbles. Does their straight line also pass through (0, 0)?

F. Green : white is 4 : 1. There are 30 beads. Five equal parts means __ beads per part. Green = __; white = __.

D. Leo says 2 yellow : 3 blue means $\frac{2}{3}$ of all the paint is yellow. Repair his statement.

M1. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

M2. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

M3. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

X1. Gold: silver beads are 3 : 5. There are 40 beads. Find each count.

X2. Orange: purple tiles are 5 : 3. What fraction of all tiles is orange?

S. To compare two mixtures fairly, I can make ___ the same.

E. Mix A has 4 cups juice and 3 cups water. Mix B has 6 cups juice and 5 cups water. Which has the larger fraction of juice?

CHALLENGE / Explain your plan and check the result

C1. Gold: silver beads are 3 : 4 in a 42-bead bracelet kit. Add 6 gold beads. Find the original counts and the new gold: silver ratio.

C2. Mix A uses 2 cups concentrate for 3 cups water. Mix B uses 3 cups concentrate for 5 cups water. Compare at 15 cups water, then say which tastes stronger.

Week 2 reference

A ratio compares two amounts in a chosen order. Equivalent ratios scale both amounts together. Tables and double number lines keep matching amounts connected.

Equivalent ratios

Equivalent ratios multiply or divide both quantities by the same factor.

Example: A mix uses 3 cups oats for every 2 cups seeds. Find seeds for 12 cups oats.

12 ÷ 3 = 4 batches.

Each batch has 2 cups seeds.

4 × 2 = 8 cups seeds.

Result: 8 cups seeds

Ratio tables and comparisons

A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.

Example: Orange : purple tiles is 3 : 2. There are 25 tiles. Find each count.

3 + 2 = 5 equal parts.

25 ÷ 5 = 5 tiles per part.

Orange: 3 × 5 = 15. Purple: 2 × 5 = 10.

Result: 15 orange, 10 purple

Words to use

Ratio: A comparison of two amounts in a stated order.

Equivalent ratios: Ratios that describe the same relationship.

Scale factor: The number used to multiply both amounts.

Ratio table: A table of matching amounts with the same ratio.

Double number line: Two aligned scales showing matching amounts.

Ratios and matching amounts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 4×3

2. 4×7

3. 4×4

4. 4×9

5. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

6. $2.5 + 0.25$

7. 5×5

8. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{9}{2}$ cups rice?

9. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

10. Is 6 : 10 equivalent to 9 : 15?

11. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

12. A team has 4 runners for every 3 walkers. There are 18 walkers. How many runners?

13. Mia changes 4 red : 3 blue to 7 red : 6 blue by adding 3 to each count. Is the ratio unchanged?

14. Red: blue is 2 : 3. Complete (2, 3), (4,__).

15. A mix has 9 raisins and 12 nuts. Divide both counts by 3; write the new ratio.

16. A recipe uses 8 cups oats for 10 cups seeds. How many cups seeds go with 12 cups oats?

17. Gold: silver beads are 5 : 6. There are 66 beads. Find each count.

18. Orange: purple tiles are 6 : 7. What fraction of all tiles is orange?

CHALLENGE / Explain your plan and check the result

C1. A banner repeats 4 red flags and 7 green flags. You have 28 red and 52 green flags. Make as many complete patterns as possible. How many flags of each color are unused?

C2. Paint uses 4 cups yellow for 6 cups blue. A small batch uses 3 cups yellow. Find the blue amount and total paint. Explain a non-whole scale factor.

Ratios and matching amounts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Compare $\frac{1}{3}$ and $\frac{1}{4}$.

6. 300 ÷ 100

7. 72 ÷ 6

8. Yellow : purple is 2 : 5. There are 28 beads. Find both counts.

9. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

10. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

11. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

12. A mix has 5 cups oats and 3 cups seeds. What fraction of the whole is seeds?

13. Leo says 2 yellow : 3 blue means $\frac{2}{3}$ of all the paint is yellow. Repair his statement.

14. Red: blue is 4 : 4. Complete (4, 4), (16,__).

15. A mix has 10 raisins and 8 nuts. Divide both counts by 2; write the new ratio.

16. A recipe uses 6 cups oats for 14 cups seeds. How many cups seeds go with 9 cups oats?

17. Gold: silver beads are 4 : 3. There are 21 beads. Find each count.

18. Orange: purple tiles are 3 : 8. What fraction of all tiles is orange?

CHALLENGE / Explain your plan and check the result

C1. A table lists (bags, marbles): (2, 14), (5, 35), (8, 54). One pair breaks the ratio. Identify it, repair it, and give the pair for zero bags.

C2. Team A has red: blue ribbons 2 : 3. Team B has 3 : 5. Each team has 40 ribbons. Find red ribbons for each team and compare the red fractions.

Ratios and matching amounts

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 4.2×10

2. $400 \div 100$

3. Double 4.5

4. Half of 24

5. Is 6 : 10 equivalent to 9 : 15?

6. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

7. Find $3 \times \frac{2}{3}$.

8. Find $\frac{1}{2} + \frac{1}{3}$.

9. $4.5 + 0.25$

10. A box is 4 cm by 3 cm by 3 cm. Find its volume.

11. 7×2

12. A box is 7 cm by 2 cm by 3 cm. Find its volume.

13. A tile bag has red : blue = 3 : 2. The total is greater than 20 and less than 30. How many of each color?

14. Red: blue is 6 : 5. Complete (6, 5), (36,__).

15. A mix has 10 raisins and 20 nuts. Divide both counts by 5; write the new ratio.

16. A recipe uses 4 cups oats for 8 cups seeds. How many cups seeds go with 6 cups oats?

17. Gold: silver beads are 3 : 5. There are 40 beads. Find each count.

18. Orange: purple tiles are 5 : 3. What fraction of all tiles is orange?

19. Complete the ratio table: (5, 7), (25,__), (__, 42).

20. Is 4 : 6 equivalent to 2 : 4? Explain without using the difference.

21. Mix A has 3 cups fruit for 5 cups yogurt. Mix B has 7 for 10. Which has more fruit for the same yogurt?

22. For each box there are 4 pencils. Write ordered pairs for 0, 4, and 5 boxes. Is (0, 0) included?

23. 8 bags each hold $\frac{3}{6}$ kilogram. Find the total mass.

CHALLENGE / Explain your plan and check the result

C1. A box has 72 beads with red: blue 4 : 5. Without adding or removing beads, recolor some blue beads red to make equal counts. How many must be recolored?

C2. The pair (n, 12) has ratio 2 : 3. Find n. Then test whether adding 6 to both entries preserves the ratio. Give a correct equivalent pair with first entry 14.

Ratios and matching amounts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 448 to the nearest hundred.

2. Round 4.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. Is 6 : 10 equivalent to 9 : 15?

6. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

7. Simplify the ratio 16 : 28 by dividing both counts by 4.

8. Write a story for $\frac{1}{6} \div 3$ and solve it.

9. Find $\frac{1}{2} + \frac{1}{3}$.

10. Find $\frac{2}{3} + \frac{1}{4}$.

11. $\frac{4}{3} - \frac{3}{4}$

12. $\frac{3}{2} - \frac{3}{4}$

13. A team has 4 runners for every 3 walkers. There are 18 walkers. How many runners?

14. Leo says 2 yellow : 3 blue means $\frac{2}{3}$ of all the paint is yellow. Repair his statement.

15. Red: blue is 3 : 6. Complete (3, 6), (9,__).

16. A mix has 16 raisins and 16 nuts. Divide both counts by 4; write the new ratio.

17. A recipe uses 10 cups oats for 12 cups seeds. How many cups seeds go with 15 cups oats?

18. Gold: silver beads are 2 : 3. There are 15 beads. Find each count.

CHALLENGE / Explain your plan and check the result

C1. A recipe uses flour: sugar 5 : 2. You have 20 cups flour and 10 cups sugar. Use all the flour. Find the sugar used, sugar left, and total ingredients in the batch.

C2. A row repeats 2 triangles then 3 squares. How many of each shape occur in the first 37 shapes? State the triangle: square ratio for those 37 shapes.

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Is 6 : 10 equivalent to 9 : 15?

2. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

3. Simplify the ratio 16 : 28 by dividing both counts by 4.

4. Simplify the ratio 8 : 14 by dividing both counts by 2.

5. Compare $\frac{2}{4}$ and $\frac{2}{5}$.

6. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

7. Find $\frac{3}{4} \times \frac{4}{5}$.

8. 6.5 + 0.25

9. A box is 6 cm by 5 cm by 3 cm. Find its volume.

10. Mia changes 4 red : 3 blue to 7 red : 6 blue by adding 3 to each count. Is the ratio unchanged?

11. Red: blue is 6 : 7. Complete (6, 7), (36,__).

12. A mix has 6 raisins and 24 nuts. Divide both counts by 3; write the new ratio.

13. A recipe uses 8 cups oats for 8 cups seeds. How many cups seeds go with 12 cups oats?

14. Gold: silver beads are 5 : 5. There are 50 beads. Find each count.

15. Orange: purple tiles are 5 : 5. What fraction of all tiles is orange?

16. Complete the ratio table: (3, 7), (9,__), (__, 28).

17. Is 6 : 8 equivalent to 4 : 6? Explain without using the difference.

18. Mix A has 3 cups fruit for 5 cups yogurt. Mix B has 7 for 10. Which has more fruit for the same yogurt?

CHALLENGE / Explain your plan and check the result

C1. Lemonade A has lemon: water 3 : 7. Lemonade B has 2 : 5. Make 35 cups water in each recipe. Which needs more lemon, and how much more?

C2. A club has adults: children 2 : 5 and 35 people. Ten more children arrive. Find the original counts, then the new adults: children ratio in simplest form.

Rates and unit rates

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Find $\frac{2}{3} + \frac{1}{4}$.

R2. Yellow : purple is 2 : 5. There are 28 beads. Find both counts.

R3. Share $\frac{1}{3}$ liter equally among 2 cups. How much per cup?

R4. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{9}{2}$ cups rice?

P1. A rider goes 20 kilometers in 4 hours at a steady speed. Find kilometers per hour.

P2. Eight apples cost $6.00. Find dollars per apple.

P3. Two cups of rice make 5 servings. Find cups per serving.

P4. A printer makes 15 cards in 3 minutes. Find cards per minute and minutes per card.

P5. Pack A has 6 pencils for $2.40. Pack B has 10 pencils for $3.50. Which costs less per pencil?

P6. A skater travels 3 miles in 20 minutes at steady speed. Find miles per hour.

P7. A robot rolls 4 meters each second. Complete distances for 0, 2, and 5 seconds.

F. A scooter travels 5 miles in 30 minutes. There are __ half-hours in one hour, so the speed is __ miles per hour.

D. Ava says 8 apples for $6 costs $\frac{4}{3}$ dollars per apple because $8 \div 6 = \frac{4}{3}$. Is her unit correct?

X1. A cyclist travels 20 kilometers in 2 hours at steady speed. Find kilometers per hour.

X2. 8 markers cost $2.8. Find dollars per marker.

X3. A printer makes 28 cards in 4 minutes. Find cards per minute and minutes per card.

S. The word per tells me to find the amount for one ___.

E. A cyclist travels 9 miles in 45 minutes at steady speed. Find miles per hour and minutes per mile.

CHALLENGE / Explain your plan and check the result

C1. Runner A travels 6 miles in 30 minutes. Runner B travels 8 miles in 40 minutes. At steady speeds, who is faster? How far does each travel in 75 minutes?

C2. Shop A sells 4 notebooks for $6. Shop B sells 6 for $8.40. Find each unit price. Buying 12 notebooks, how much does the cheaper choice save?

Percent as a rate per hundred

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Write a story for $\frac{1}{6} \div 3$ and solve it.

R2. A team has 4 runners for every 3 walkers. There are 18 walkers. How many runners?

R3. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

R4. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

P1. Find 10%, 25%, 50%, and 75% of 80 counters.

P2. A shelf holds 40 books. Twelve are mysteries. What percent are mysteries?

P3. Twenty-four stickers are 40% of a collection. How many stickers are in the whole collection?

P4. Find 125% of 48 minutes.

P5. Find 35% of 60 tiles.

P6. Find 15% of $18.00.

F. Twelve coins are 20% of a jar. Ten percent is __ coins. One hundred percent is __ coins.

D. Omar says 24 is 40% of 9.6 because $0.40 \times 24 = 9.6$. Explain the error.

X1. Find 35% of 140. Name the whole.

X2. 20 stickers are 25% of a collection. How many stickers are in the whole collection?

X3. 3 of 20 books are mysteries. What percent are mysteries?

X4. Find 10% of $13. Give a money answer.

S. Before using a percent, I decide which amount represents ___.

E. Twenty-one marbles are 35% of a bag. How many marbles are in the bag?

CHALLENGE / Explain your plan and check the result

C1. Twenty-four fiction books are 30% of a shelf. The shelf also has science books equal to 25% of all books. Find the total and science count. Which group has more books?

C2. Team A collects 125% of its 48-can goal. Team B collects 150% of its 40-can goal. Find both totals and compare them. Explain why the larger percent need not give more cans.

Mini Check A

Tutor choice: use this check in place of the routine questions, or in a separate assessment sitting. Keep the two daily challenges for discussion.

1. Compare $\frac{2}{9}$ and $\frac{7}{9}$.

2. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

3. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

4. Is 6 : 10 equivalent to 9 : 15?

5. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

6. Pack A has 4 erasers for $1.80. Pack B has 6 for $2.40. Which costs less per eraser?

7. Find 140% of 30 points.

8. Find 35% of 80 tiles.

Week 3 reference

A unit rate describes an amount per one unit. Percent means per hundred. Label the whole before deciding whether to find a part, a whole, or a percent.

Rates and unit rates

A unit rate names how much of one quantity goes with one unit of another. Its units show the division order.

Example: A walker travels 2 miles in 30 minutes. Find miles per hour.

60 minutes is two groups of 30.

Two groups of 2 miles make 4 miles.

The rate is 4 miles per hour.

Result: 4 miles per hour

Percent as a rate per hundred

Percent compares an amount with a whole split into 100 equal parts.

Example: Eighteen blocks are 30% of a tower set. Find the whole set.

30% has three 10% groups.

18 ÷ 3 = 6 blocks per 10%.

Ten groups give 10 × 6 = 60 blocks.

Result: 60 blocks

Words to use

Rate: A ratio comparing quantities with different units.

Unit rate: An amount per one unit of another quantity.

Per: For each one.

Percent: A rate per hundred.

Whole: The amount that represents 100%.

Rates and percents

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 5×3

2. 5×7

3. 5×4

4. 5×9

5. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

6. $3.5 + 0.25$

7. 6×6

8. A walker travels 12 miles in 3 hours. Find miles per hour.

9. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

10. Pack A has 4 erasers for $1.80. Pack B has 6 for $2.40. Which costs less per eraser?

11. A machine makes 24 labels in 6 minutes. Find labels per minute and minutes per label.

12. A snail moves 3 centimeters each minute. Find the distance after 0, 4, and 7 minutes.

13. Ava says 8 apples for $6 costs $\frac{4}{3}$ dollars per apple because $8 \div 6 = \frac{4}{3}$. Is her unit correct?

14. A cyclist travels 16 kilometers in 2 hours at steady speed. Find kilometers per hour.

15. 4 markers cost $2.6. Find dollars per marker.

16. A printer makes 18 cards in 6 minutes. Find cards per minute and minutes per card.

17. A runner goes 3 miles in 20 minutes at steady speed. Find miles per hour.

18. Pack A: 7 pencils for $4.2. Pack B: 9 for $5.85. Which costs less per pencil?

CHALLENGE / Explain your plan and check the result

C1. A machine prints 8 labels in 3 minutes. Find labels per minute and minutes per label. How long will 40 labels take at that rate?

C2. A cyclist goes 9 miles in 45 minutes at steady speed. If the ride continues 30 more minutes at that speed, find the extra distance and the entire distance.

Rates and percents

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. A mix has 5 cups oats and 3 cups seeds. What fraction of the whole is seeds?

6. A box is 4 cm by 3 cm by 3 cm. Find its volume.

7. 84 ÷ 7

8. Find 10%, 50%, and 75% of 40 counters.

9. Find 140% of 30 points.

10. Find 35% of 80 tiles.

11. 12 shells are 30% of a collection. Find the whole collection.

12. 9 of 12 tiles are green. What percent are green?

13. Omar says 24 is 40% of 9.6 because 0.40 × 24 = 9.6. Explain the error.

14. Find 30% of 100. Name the whole.

15. 120 stickers are 75% of a collection. How many stickers are in the whole collection?

16. 9 of 60 books are mysteries. What percent are mysteries?

17. Find 5% of $27. Give a money answer.

18. A cyclist travels 9 kilometers in 3 hours at steady speed. Find kilometers per hour.

CHALLENGE / Explain your plan and check the result

C1. A $45 bag is reduced by 20% of its price. A $48 bag is reduced by 25% of its price. Find both reductions and final prices. Which final price is lower?

C2. A class has 32 students. Twelve ride bicycles and 8 walk. Find the percent in each group and the percent who do neither. Assume the groups do not overlap.

Rates and percents

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 5.2 × 10

2. 500 ÷ 100

3. Double 5.5

4. Half of 30

5. A walker travels 12 miles in 3 hours. Find miles per hour.

6. 9 of 12 tiles are green. What percent are green?

7. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

8. A line marks 0, 3, and 6. How far from 3 toward 6 is $4\frac{1}{2}$?

9. $\frac{1}{4} + \frac{1}{4}$

10. 500 ÷ 100

11. 8 × 3

12. A box is 8 cm by 3 cm by 3 cm. Find its volume.

13. A mystery number has this property: 25% of it is 6 more than 10% of it. Find the number.

14. A cyclist travels 36 kilometers in 6 hours at steady speed. Find kilometers per hour.

15. 6 markers cost $3.3. Find dollars per marker.

16. A printer makes 20 cards in 4 minutes. Find cards per minute and minutes per card.

17. A runner goes 3 miles in 15 minutes at steady speed. Find miles per hour.

18. Pack A: 6 pencils for $3.3. Pack B: 8 for $4.8. Which costs less per pencil?

19. A robot rolls 3 meters per second. Complete distances for 0, 2, and 4 seconds.

20. Find 20% of 40. Name the whole.

21. 20 stickers are 25% of a collection. How many stickers are in the whole collection?

22. 28 of 80 books are mysteries. What percent are mysteries?

23. Find 25% of $16. Give a money answer.

CHALLENGE / Explain your plan and check the result

C1. Flowers occupy 40% of a garden and cover 24 square meters. Beans use $\frac{2}{3}$ of the remaining area. Find the whole area, bean area, and area left for paths.

C2. Cyclist A goes 10 km in 40 minutes; B goes 9 km in 30 minutes. Find both hourly rates. How long would B need for 12 km?

Rates and percents

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 548 to the nearest hundred.

2. Round 5.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. A walker travels 12 miles in 3 hours. Find miles per hour.

6. 9 of 12 tiles are green. What percent are green?

7. 12 shells are 30% of a collection. Find the whole collection.

8. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

9. 2.4 + 1.75

10. 3.4 × 0.5

11. 4.4 + 1.75

12. 5.4 × 0.5

13. A walker travels 16 miles in 4 hours. Find miles per hour.

14. Omar says 24 is 40% of 9.6 because 0.40 × 24 = 9.6. Explain the error.

15. A cyclist travels 20 kilometers in 4 hours at steady speed. Find kilometers per hour.

16. 7 markers cost $3.5. Find dollars per marker.

17. A printer makes 18 cards in 3 minutes. Find cards per minute and minutes per card.

18. A runner goes 7 miles in 10 minutes at steady speed. Find miles per hour.

CHALLENGE / Explain your plan and check the result

C1. You have $15 and reserve 20% for a bus fare. Pens cost $1.50 each. What is the reserved amount, and what is the greatest number of pens you can buy with the rest?

C2. A tank contains 18 liters, which is 30% of its capacity. Water flows in at 6 liters per minute. How many liters are needed to fill it, and how long will that take?

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. A walker travels 12 miles in 3 hours. Find miles per hour.

2. 9 of 12 tiles are green. What percent are green?

3. 12 shells are 30% of a collection. Find the whole collection.

4. A walker travels 8 miles in 2 hours. Find miles per hour.

5. Is $\frac{2}{3} \times 9$ smaller or larger than 9? Compute to check.

6. How many $\frac{1}{4}$-meter pieces fit in 3 meters?

7. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

8. $\frac{1}{6} + \frac{1}{6}$

9. $700 \div 100$

10. Ava says 8 apples for \$6 costs $\frac{4}{3}$ dollars per apple because $8 \div 6 = \frac{4}{3}$. Is her unit correct?

11. A cyclist travels 24 kilometers in 6 hours at steady speed. Find kilometers per hour.

12. 8 markers cost \$3.6. Find dollars per marker.

13. A printer makes 42 cards in 6 minutes. Find cards per minute and minutes per card.

14. A runner goes 7 miles in 15 minutes at steady speed. Find miles per hour.

15. Pack A: 6 pencils for \$3.3. Pack B: 8 for \$4.8. Which costs less per pencil?

16. A robot rolls 5 meters per second. Complete distances for 0, 6, and 8 seconds.

17. Find 60% of 160. Name the whole.

18. 20 stickers are 25% of a collection. How many stickers are in the whole collection?

CHALLENGE / Explain your plan and check the result

C1. A trip includes 15% of a 200-student school. The trip needs 2 adult guides for every 15 students. Find the participating students and required adults.

C2. A basket holds 30 oranges. Some are used; the 12 remaining are 40% of the original count. Check this claim, then find what percent was used.

The meaning of fraction division

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

R2. Red : blue is 3 : 5. How many blue tiles go with 12 red tiles?

R3. 5 markers cost $3.25. Find dollars per marker.

R4. Find both $4 \div \frac{1}{5}$ and $\frac{1}{5} \div 4$.

P1. Four meters are cut into $\frac{1}{5}$-meter pieces. How many pieces?

P2. How many $\frac{2}{3}$-cup scoops fit in 4 cups?

P3. How many $\frac{1}{8}$-meter pieces fit in $\frac{3}{4}$ meter?

P4. How many $\frac{2}{3}$-cup groups fit in $\frac{5}{6}$ cup? A fraction of a group is allowed.

P5. What fraction of a $\frac{5}{6}$-cup scoop is filled by $\frac{1}{3}$ cup?

P6. A jar is $\frac{2}{3}$ full after $\frac{3}{5}$ liter is poured in. How many liters fill the whole jar?

P7. Write a story for $\frac{3}{4} \div \frac{1}{2}$. Solve it.

F. Find $\frac{5}{8} \div \frac{1}{4}$. One group is __ eighths. Five eighths holds __ groups.

D. Ben says $\frac{1}{3} \div \frac{5}{6} = \frac{5}{18}$. He multiplies the fractions. Explain what division asks here.

X1. How many $\frac{1}{4}$-sized groups fit in $\frac{7}{8}$? Include a part of a group.

X2. How many $\frac{3}{4}$-cup scoops fit in 5 cups? A partial scoop is allowed.

X3. Share $\frac{3}{7}$ liter equally among 4 bottles. Find each share.

S. To divide by a fraction, I ask how many ___-sized groups fit in ___.

E. How much of a $\frac{3}{4}$-meter length is $\frac{2}{5}$ meter? Draw equal parts to show $\frac{2}{5} \div \frac{3}{4}$.

CHALLENGE / Explain your plan and check the result

C1. A 2 $\frac{1}{2}$-meter ribbon is cut into $\frac{3}{4}$-meter pieces. How many full pieces can you cut? How much ribbon remains? Explain why the quotient is not the full-piece count.

C2. Pitcher A holds $\frac{3}{4}$ liter and fills $\frac{3}{8}$-liter cups. Pitcher B holds 1 liter and fills $\frac{5}{12}$-liter cups. Compare exact cup counts and full cups.

Fraction division and reciprocals

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Find $\frac{1}{2} + \frac{1}{3}$.

R2. Complete the ratio-table pair: (3, 4), (9, __).

R3. 9 of 12 tiles are green. What percent are green?

R4. How many $\frac{1}{4}$-meter pieces fit in 3 meters?

P1. Share $\frac{3}{4}$ liter equally among 5 bottles. How much per bottle?

P2. How many $\frac{3}{5}$-cup groups fit in 4 cups? Include a fraction of a group.

P3. Find $\frac{7}{8} \div \frac{1}{4}$. Check by multiplying the quotient by $\frac{1}{4}$.

P4. Find $1\frac{7}{8} \div \frac{5}{8}$.

P5. A cord is $3\frac{1}{2}$ meters long. Each piece is $\frac{7}{8}$ meter. How many pieces?

P6. A rectangle has area $\frac{5}{6}$ square meter and width $\frac{2}{3}$ meter. Find its length.

P7. A box has edges $\frac{2}{3}$, $\frac{3}{4}$, and 2 meters. Find its volume.

F. Complete: $\frac{4}{5} \div \frac{2}{3} = \frac{4}{5} \times __ = __$. Check by multiplying by $\frac{2}{3}$.

D. Zoe writes $\frac{7}{8} \div \frac{1}{4} = \frac{8}{7} \times \frac{1}{4} = \frac{2}{7}$. Explain which number should be inverted.

M1. Find $3 \times \frac{2}{3}$.

M2. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{9}{2}$ cups rice?

M3. 3 markers cost $1.95. Find dollars per marker.

S. I can check a fraction quotient by multiplying it by ___.

E. A ribbon is $2\frac{5}{8}$ yards long. Each piece is $\frac{3}{8}$ yard. How many pieces can be cut?

CHALLENGE / Explain your plan and check the result

C1. A rectangular card has area $\frac{5}{6}$ square meter and width $\frac{2}{3}$ meter. Find its length, then the distance around its edge.

C2. A box measures $\frac{3}{4}$ meter by $\frac{2}{3}$ meter by $2\frac{1}{2}$ meters. Find its volume. Two such boxes occupy a 3-cubic-meter space; how much volume is unoccupied?

Week 4 reference

Division asks how many divisor-sized groups fit. Equal small units make the comparison visible. The reciprocal rule records that reasoning efficiently.

The meaning of fraction division

The divisor names one group. A quotient tells how many of those groups fit, even when it is part of a group.

Example: Find $\frac{3}{4} \div \frac{1}{2}$ with a tape.

Rename one half as two fourths.

Three fourths contains one full pair of fourths and one fourth left.

The remaining fourth is half of a two-fourths group.

Result: 1 $\frac{1}{2}$

Fraction division and reciprocals

Dividing by a fraction counts groups. Multiplying by its reciprocal gives the same count.

Example: Find $\frac{2}{3} \div \frac{3}{5}$.

One whole contains $\frac{5}{3}$ groups of size $\frac{3}{5}$.

Two thirds of a whole contains $\frac{2}{3} \times \frac{5}{3}$ groups.

Multiply to get $\frac{10}{9}$, or 1 $\frac{1}{9}$.

Result: 1 $\frac{1}{9}$

Words to use

Dividend: The amount being divided.

Divisor: The amount that names one group in a grouping problem.

Quotient: The result of division.

Reciprocal: A number that multiplies with another number to make 1.

Mixed number: A whole number plus a fraction.

Dividing fractions

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 6×3

2. 6×7

3. 6×4

4. 6×9

5. Find 10%, 50%, and 75% of 40 counters.

6. Find both $4 \div \frac{1}{5}$ and $\frac{1}{5} \div 4$.

7. 7×2

8. Find $\frac{1}{2} \div \frac{3}{4}$.

9. Find $\frac{1}{2} \div \frac{4}{5}$.

10. $\frac{2}{3} \div \frac{1}{5}$

11. $1 \div \frac{3}{5}$

12. $\frac{5}{3} \div \frac{1}{5}$

13. Ben says $\frac{1}{3} \div \frac{5}{6} = \frac{5}{18}$. He multiplies the fractions. Explain what division asks here.

14. How many $\frac{2}{3}$-sized groups fit in $\frac{5}{6}$? Include a part of a group.

15. How many $\frac{3}{4}$-cup scoops fit in 7 cups? A partial scoop is allowed.

16. Share $\frac{2}{5}$ liter equally among 4 bottles. Find each share.

17. 4 bags each hold $\frac{2}{5}$ kilogram. Find the total mass.

18. Find $\frac{7}{8}$ of $\frac{6}{10}$. Sketch equal parts.

CHALLENGE / Explain your plan and check the result

C1. A jar contains $\frac{3}{5}$ liter and is $\frac{2}{3}$ full. Find its full capacity and the extra liquid needed to fill it.

C2. A $\frac{3}{4}$-batch of dough uses $\frac{4}{5}$ cup of oil. How much oil does one full batch need? How much for $1\ \frac{1}{2}$ batches?

Dividing fractions

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find 140% of 30 points.

6. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

7. $96 \div 8$

8. Find $3\ \frac{1}{2} \div \frac{3}{4}$.

9. Which is the reciprocal of $2\ \frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

10. Find $1\ \frac{1}{2} \div \frac{3}{4}$.

11. $\frac{4}{3} \div \frac{3}{5}$

12. $\frac{4}{3} \times \frac{4}{5}$

13. Zoe writes $\frac{7}{8} \div \frac{1}{4} = \frac{8}{7} \times \frac{1}{4} = \frac{2}{7}$. Explain which number should be inverted.

14. How many $\frac{3}{5}$-sized groups fit in $\frac{9}{10}$? Include a part of a group.

15. How many $\frac{5}{6}$-cup scoops fit in 2 cups? A partial scoop is allowed.

16. Share $\frac{3}{8}$ liter equally among 6 bottles. Find each share.

17. Find $3\ \frac{1}{6} \div \frac{4}{5}$. Check by multiplication.

18. A rectangle has area $\frac{2}{5}$ square meter and width $\frac{2}{5}$ meter. Find its length.

CHALLENGE / Explain your plan and check the result

C1. There are $3\ \frac{1}{2}$ kilograms of rice. Each meal uses $\frac{2}{5}$ kilogram. Find the greatest number of complete meals and the rice left over.

C2. A poster has area $1\ \frac{7}{8}$ square meters and width $\frac{3}{4}$ meter. Another poster has the same area and width $\frac{5}{8}$ meter. Find both lengths. Which must be longer, and why?

Dividing fractions

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 6.2×10

2. $600 \div 100$

3. Double 6.5

4. Half of 36

5. Find $3 \div \frac{2}{3}$.

6. Find $\frac{1}{2} \div \frac{4}{5}$.

7. Complete the ratio-table pair: (3, 4), (12, __).

8. A recipe uses 2 cups beans for 3 cups rice. How many cups beans go with $\frac{15}{2}$ cups rice?

9. Find $\frac{1}{2} + \frac{1}{3}$.

10. Find $\frac{3}{4} \times \frac{4}{5}$.

11. 2×4

12. A box is 2 cm by 4 cm by 3 cm. Find its volume.

13. A jug holds 2 liters. You have $\frac{1}{3}$-liter and $\frac{1}{2}$-liter scoops. Use both sizes and exactly five full scoops to fill it. How many of each?

14. How many $\frac{1}{2}$-sized groups fit in $\frac{3}{4}$? Include a part of a group.

15. How many $\frac{3}{4}$-cup scoops fit in 3 cups? A partial scoop is allowed.

16. Share $\frac{4}{7}$ liter equally among 4 bottles. Find each share.

17. Find $1\frac{1}{4} \div \frac{2}{5}$. Check by multiplication.

18. A rectangle has area $\frac{21}{20}$ square meter and width $\frac{3}{5}$ meter. Find its length.

19. A box has edges $\frac{4}{3}$, $\frac{3}{4}$, and 4 meters. Find its volume.

20. A cyclist travels 27 kilometers in 3 hours at steady speed. Find kilometers per hour.

21. 5 markers cost $3.5. Find dollars per marker.

22. A printer makes 12 cards in 3 minutes. Find cards per minute and minutes per card.

23. A runner goes 4 miles in 30 minutes at steady speed. Find miles per hour.

CHALLENGE / Explain your plan and check the result

C1. A bottle holds $\frac{7}{8}$ liter. You pour out $\frac{1}{4}$ liter, then share the remainder equally among 5 cups. Find each share and check that all liquid is accounted for.

C2. A $\frac{4}{5}$-meter ribbon is $2\frac{2}{3}$ times one piece's length. Find one piece's length. Then find the exact number of piece-lengths in 2 meters and the number of complete pieces.

Dividing fractions

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 648 to the nearest hundred.

2. Round 6.72 to the nearest whole number.

3. Use 20×5 to estimate 19.8×5.1.

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

5. Find $3 \div \frac{2}{3}$.

6. Find $\frac{1}{2} \div \frac{4}{5}$.

7. Jules says $\frac{1}{2} \div \frac{2}{3} = \frac{1}{3}$. Find and fix the error.

8. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

9. Find 12% of $15.00.

10. 18 shells are 30% of a collection. Find the whole collection.

11. 12 of 16 tiles are green. What percent are green?

12. Find 25% of 40 stickers.

13. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

14. Zoe writes $\frac{7}{8} \div \frac{1}{4} = \frac{8}{7} \times \frac{1}{4} = \frac{2}{7}$. Explain which number should be inverted.

15. How many $\frac{1}{3}$-sized groups fit in $\frac{7}{12}$? Include a part of a group.

16. How many $\frac{5}{6}$-cup scoops fit in 4 cups? A partial scoop is allowed.

17. Share $\frac{2}{6}$ liter equally among 6 bottles. Find each share.

18. Find $2\frac{1}{5} \div \frac{3}{5}$. Check by multiplication.

CHALLENGE / Explain your plan and check the result

C1. A tailor has 5 meters of cloth and sets aside $\frac{1}{2}$ meter. Each bag needs $\frac{3}{4}$ meter. How many bags can be made? If one more bag is ordered, how much extra cloth is needed?

C2. A cuboid has volume $\frac{9}{8}$ cubic meter. Its base is $\frac{3}{4}$ meter by $\frac{1}{2}$ meter. Find the height. Then find the volume of a cuboid with the same base and half that height.

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Find $3 \div \frac{2}{3}$.

2. Find $\frac{1}{2} \div \frac{4}{5}$.

3. Jules says $\frac{1}{2} \div \frac{2}{3} = \frac{1}{3}$. Find and fix the error.

4. A rectangle has area $\frac{5}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

5. 3 markers cost $1.95. Find dollars per marker.

6. Find 10%, 50%, and 75% of 40 counters.

7. Complete the ratio-table pair: (3, 4), (15, __).

8. Find $\frac{1}{2} + \frac{1}{3}$.

9. Find $\frac{3}{4} \times \frac{4}{5}$.

10. Ben says $\frac{1}{3} \div \frac{5}{6} = \frac{5}{18}$. He multiplies the fractions. Explain what division asks here.

11. How many $\frac{9}{10}$-sized groups fit in $\frac{3}{5}$? Include a part of a group.

12. How many $\frac{2}{3}$-cup scoops fit in 5 cups? A partial scoop is allowed.

13. Share $\frac{3}{5}$ liter equally among 3 bottles. Find each share.

14. Find $3\frac{1}{6} \div \frac{4}{5}$. Check by multiplication.

15. A rectangle has area $\frac{3}{5}$ square meter and width $\frac{2}{5}$ meter. Find its length.

16. A box has edges 1, $\frac{1}{2}$, and 3 meters. Find its volume.

17. Find 25% of 100. Name the whole.

18. 4 stickers are 10% of a collection. How many stickers are in the whole collection?

CHALLENGE / Explain your plan and check the result

C1. Two groups share juice. Group A shares $\frac{5}{6}$ liter among 5 people; Group B shares $\frac{3}{4}$ liter among 4 people. Find each person’s share and which group receives more per person.

C2. A student claims $\frac{2}{3} \div \frac{4}{5} = \frac{8}{15}$. Check that answer by multiplication, then find the correct quotient and explain why it is less than 1.

Decimal place value and products

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Complete the ratio-table pair: (3, 4), (12, __).

R2. 18 shells are 30% of a collection. Find the whole collection.

R3. A bottle is $\frac{3}{4}$ full and holds $\frac{2}{3}$ liter so far. Find its full capacity.

R4. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

P1. Find 4.7 + 8.036.

P2. Find 12 − 4.685.

P3. A rope costs $2.35 per meter. What do 4 meters cost?

P4. Find 0.6 × 0.4. Is the product less than each factor?

P5. Find 0.15 × 0.02.

P6. A notebook costs $3.85 and a pen costs $1.67. You pay $10. Find the change.

F. Complete: 0.3 × 0.07 means 3 tenths × 7 hundredths = 21 __ = __.

D. Isaac says 0.6 × 0.4 = 2.4 because 6 × 4 = 24. Use place value to repair this.

X1. Find 2.7 + 0.154. Align equal place values.

X2. Find 12 − 0.181. Show regrouping across zeros.

X3. Find 0.6 × 0.05. Explain the product's place value.

X4. Ribbon costs $1.85 per meter. What do 7 meters cost?

S. I check a decimal answer by naming the units and ___.

E. Find 0.45 × 0.06. Explain why your decimal position is reasonable.

CHALLENGE / Explain your plan and check the result

C1. Fabric costs $3.75 per meter. Buy 2.4 meters and pay $20. Find the cost and change. Use an estimate to check the decimal placement.

C2. Rectangle A is 0.4 meter by 0.35 meter. Rectangle B is 0.14 meter by 0.9 meter. Find both areas and their difference. Explain why B’s area is less than 0.14 square meter.

Long division with decimals

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Red : blue is 3 : 5. How many blue tiles go with 6 red tiles?

R2. A snail moves 3 centimeters each minute. Find the distance after 0, 4, and 7 minutes.

R3. Find $\frac{3}{5} \div \frac{1}{5}$.

R4. Complete the ratio-table pair: (3, 4), (12, __).

P1. Use long division to find 672 ÷ 24.

P2. Find 27 ÷ 8. Continue into decimal places.

P3. Find 8.4 ÷ 7.

P4. Find 6.3 ÷ 0.09 by writing an equivalent division with a whole-number divisor.

P5. Find 7 ÷ 20.

P6. A 3.6-meter ribbon is cut into 0.4-meter pieces. How many pieces?

P7. Find 1,176 ÷ 28 and check by multiplication.

F. Complete: 5.4 ÷ 0.03 = __ ÷ 3 = __.

D. Lina changes 6.3 ÷ 0.09 into 6.3 ÷ 9 and gets 0.7. Which change is missing?

M1. A mix has 5 cups oats and 3 cups seeds. What fraction of the whole is seeds?

M2. Find 12% of $15.00.

M3. Find $2\frac{1}{2} \div \frac{3}{4}$.

S. To keep a quotient unchanged, I multiply both ___ and ___ by the same number.

E. Find 8.4 ÷ 0.07. Write an equivalent division and check by multiplication.

CHALLENGE / Explain your plan and check the result

C1. A 4.8-liter container fills bottles that hold 0.15 liter each. Bottles are packed 6 to a box. How many full boxes can be made, and how many filled bottles remain?

C2. A 7.35-meter roll of tape is cut into 0.35-meter strips. Nineteen strips are used. Find the original number of strips, the remaining length, and how many unused strips remain.

Week 5 reference

Decimal digits name parts of a whole. Addition and subtraction combine equal place-value units. Division preserves its quotient when both quantities scale equally.

Decimal place value and products

Addition combines like-sized units. Multiplication combines both the number of parts and the size of each part.

Example: Find 3.8 + 0.475.

Write 3.800 + 0.475.

Add thousandths to thousandths and tenths to tenths.

3.800 + 0.475 = 4.275. The sum is a little above 4.

Result: 4.275

Long division with decimals

Multiplying the dividend and divisor by the same power of ten preserves the quotient.

Example: Find 4.8 ÷ 0.06.

Rename both quantities in hundredths.

4.8 is 480 hundredths; 0.06 is 6 hundredths.

480 ÷ 6 = 80. Check 80 × 0.06 = 4.8.

Result: 80

Words to use

Place value: A digit’s value depends on its position.

Estimate: A nearby value that helps check an answer.

Thousandth: One of 1,000 equal parts of a whole.

Remainder: The amount left after equal groups are made.

Equivalent division: A new dividend and divisor with the same quotient.

Decimals and long division

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 7×3

2. 7×7

3. 7×4

4. 7×9

5. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

6. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

7. 8×3

8. Find $1.45 + 3.078$.

9. Find $2.45 + 4.078$.

10. Find 0.25×0.02.

11. Find $10 - 2.286$.

12. Find 0.8×0.7. Is the product below both factors?

13. Isaac says $0.6 \times 0.4 = 2.4$ because $6 \times 4 = 24$. Use place value to repair this.

14. Find $6.9 + 0.392$. Align equal place values.

15. Find $15 - 0.587$. Show regrouping across zeros.

16. Find 0.6×0.03. Explain the product's place value.

17. Ribbon costs $3.95 per meter. What do 3 meters cost?

18. A notebook costs $3.44 and a pen $2.18. Pay $13. Find the change.

CHALLENGE / Explain your plan and check the result

C1. Three notebooks cost $2.65 each. Add one $1.85 pen and pay $10. Is the payment enough? Find the exact change or shortfall.

C2. A student says $0.25 \times 0.12 = 0.3$ because $25 \times 12 = 300$. Repair the answer using both place value and a quarter-of-an-amount model. Then find 0.25×1.2.

Decimals and long division

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. 18 shells are 30% of a collection. Find the whole collection.

6. Find $\frac{3}{4} - \frac{1}{3}$.

7. 24 ÷ 2

8. Find 1.4 ÷ 0.07.

9. Find 312 ÷ 24.

10. 117 × 12

11. Find 1,344 ÷ 32. Verify by multiplication.

12. 52 ÷ 4

13. Lina changes 6.3 ÷ 0.09 into 6.3 ÷ 9 and gets 0.7. Which change is missing?

14. Find 3.04 ÷ 0.04. Write an equivalent whole-number division.

15. Find 82 ÷ 25. Continue to an exact decimal.

16. Find 96 ÷ 12. Check by multiplication.

17. 5 markers cost $2. Find dollars per marker.

18. Find 3 + 0.171. Align equal place values.

CHALLENGE / Explain your plan and check the result

C1. A machine cuts 9.6 meters of wire into 0.24-meter pieces. Another machine makes 0.3-meter pieces from the same length. Find both counts and explain which count is larger.

C2. A cyclist covers 18.9 kilometers in 0.7 hour. Find kilometers per hour. At the same rate, how far does the cyclist travel in 1.2 hours?

Decimals and long division

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 7.2 × 10

2. 700 ÷ 100

3. Double 7.5

4. Half of 42

5. Find 2.1 ÷ 0.07.

6. Find 0.25 × 0.03.

7. Find 12% of $15.00.

8. 24 shells are 30% of a collection. Find the whole collection.

9. Find $\frac{3}{4} - \frac{1}{3}$.

10. Simplify the ratio 8 : 14 by dividing both counts by 2.

11. 3 × 5

12. A box is 3 cm by 5 cm by 3 cm. Find its volume.

13. Use digits 2, 4, and 8 once each in the form A.BC. Which number is closest to 5?

14. Find 3.9 + 0.222. Align equal place values.

15. Find 16 − 0.297. Show regrouping across zeros.

16. Find 0.3 × 0.09. Explain the product's place value.

17. Ribbon costs $2.45 per meter. What do 5 meters cost?

18. A notebook costs $2.34 and a pen $1.48. Pay $11. Find the change.

19. Find 2.4 ÷ 0.06. Write an equivalent whole-number division.

20. Find 46 ÷ 25. Continue to an exact decimal.

21. Find 480 ÷ 24. Check by multiplication.

22. 5 markers cost $3. Find dollars per marker.

23. How many $\frac{9}{10}$-sized groups fit in $\frac{3}{5}$? Include a part of a group.

CHALLENGE / Explain your plan and check the result

C1. A shop has 12 kilograms of rice. After 1.8 kilograms are sold, the rest is packed into 0.45-kilogram bags. Find the number of full bags and rice left unpacked.

C2. A 4.5-meter board costs $13.50. A 3.2-meter board costs $10.24. Compare prices per meter. How much would 4 meters cost at each unit price?

Decimals and long division

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 748 to the nearest hundred.

2. Round 7.72 to the nearest whole number.

3. Use 20×5 to estimate 19.8×5.1.

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

5. Find $2.1 \div 0.07$.

6. Find 0.25×0.03.

7. Find $7 \div 8$.

8. Find $\frac{2}{4} \div \frac{1}{4}$.

9. Find both $4 \div \frac{1}{5}$ and $\frac{1}{5} \div 4$.

10. Find $4 \times \frac{2}{3}$.

11. $\frac{2}{3} \times \frac{2}{5}$

12. $2 \times \frac{2}{5}$

13. A ball costs $4.65 and a card costs $2.78. Find the change from $10.

14. Lina changes $6.3 \div 0.09$ into $6.3 \div 9$ and gets 0.7. Which change is missing?

15. Find $6 + 0.341$. Align equal place values.

16. Find $12 - 0.5$. Show regrouping across zeros.

17. Find 0.3×0.08. Explain the product's place value.

18. Ribbon costs $3.50 per meter. What do 6 meters cost?

CHALLENGE / Explain your plan and check the result

C1. A fundraiser sells 24 tickets at $3.75 each. Supplies cost $28.60. What money remains? If the goal is $75 after costs, how much more is needed?

C2. A rectangular floor is 3.6 meters by 2.5 meters. Each tile covers 0.09 square meter. Ignore cutting and gaps. Find the floor area and the number of tiles needed.

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Find $2.1 \div 0.07$.

2. Find 0.25×0.03.

3. Find $7 \div 8$.

4. Find 0.25×0.04.

5. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

6. A machine makes 24 labels in 6 minutes. Find labels per minute and minutes per label.

7. Jules says $\frac{1}{2} \div \frac{2}{3} = \frac{1}{3}$. Find and fix the error.

8. Find $\frac{3}{4} - \frac{1}{3}$.

9. Simplify the ratio 8 : 14 by dividing both counts by 2.

10. Isaac says $0.6 \times 0.4 = 2.4$ because $6 \times 4 = 24$. Use place value to repair this.

11. Find $8.1 + 0.46$. Align equal place values.

12. Find $19 - 0.703$. Show regrouping across zeros.

13. Find 0.3×0.07. Explain the product's place value.

14. Ribbon costs $4.55 per meter. What do 7 meters cost?

15. A notebook costs $3.88 and a pen $2.46. Pay $13. Find the change.

16. Find $0.3 \div 0.03$. Write an equivalent whole-number division.

17. Find $16 \div 5$. Continue to an exact decimal.

18. Find $140 \div 14$. Check by multiplication.

CHALLENGE / Explain your plan and check the result

C1. Two expressions are $0.84 \div 0.07$ and $8.4 \div 0.7$. Predict whether they are equal, then calculate. Now compare $0.84 \div 0.7$ and explain the factor-of-ten change.

C2. A baker uses 0.18 kilogram of flour per loaf and makes 35 loaves from a 7-kilogram bag. Find the flour left. Can the remainder make 4 more complete loaves? Explain.

Expressions and equivalent forms

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. 3 markers cost $1.95. Find dollars per marker.

R2. Find $4 \div \frac{2}{3}$.

R3. Find 288 ÷ 24.

R4. 18 shells are 30% of a collection. Find the whole collection.

P1. A sticker sheet costs n dollars. Write the cost of 5 sheets. Find it when n = 2 and when n = 4.

P2. Find 2a + 5 when a = 6. Write the multiplication before adding.

P3. Find 4^2, 4×2, 3^3, and $(\frac{2}{3})^2$. Which pair has different meanings?

P4. A rectangle is 4 units high and x + 3 units wide. Split the width into x and 3. Write its area two ways.

P5. Simplify x + x + x + 2x. Then check your result when x = 3.

P6. Are 4(t + 2) and 4t + 2 equivalent? Check t = 1. Write a form that is equivalent to 4(t + 2).

F. Complete: 2(y + 6) = 2y + __. Check when y = 5.

D. Lena wrote 5(z + 2) = 5z + 2. Find the mistake and repair it.

X1. A packet contains 3 cards. Write the number in p packets plus 2 loose cards; evaluate at p = 2.

X2. Evaluate 2t + 3 when t = 2. Show the multiplication first.

X3. Evaluate $2^2 + 3$. Explain what the exponent means.

X4. Expand 2(x + 3) and check at x = 2.

S. Distributing works because each group contains __ and __.

E. Which expression equals 3(2x + 4) for every x? Choose 6x + 12, 6x + 4, or 5x + 7. Show how the 3 acts on both parts.

CHALLENGE / Explain your plan and check the result

C1. A school orders 6 kits. Each kit has x pencils and 4 markers. Write the total number of tools two ways. For x = 7, find the total. Explain why 6x + 4 is incorrect.

C2. Two rectangular banners are both 3 m high. Their widths are x m and x + 2 m. Write their combined area in a simplified form. Find it for x = 4. Can one rectangle show the same area?

Equations and related quantities

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Find 140% of 30 points.

R2. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

R3. Find 2.8 ÷ 0.07.

R4. A snail moves 3 centimeters each minute. Find the distance after 0, 4, and 7 minutes.

P1. Solve x + 7 = 19. Show the same change on both sides.

P2. Four identical bags hold 28 marbles. Write and solve an equation for marbles per bag, b.

P3. Solve 1.2x = 5.4. Check with multiplication.

P4. Solve $(\frac{2}{3})$ x = 10. Draw three equal parts of x first.

P5. A game costs $18. Mina has $6.50 saved. Write and solve an equation for the extra amount, a, she needs.

P6. Notebooks cost $2.50 each. Use c = 2.5n for n = 0, 1, 2, 4. Plot (n, c). Name the input and output.

F. Complete: 5x = 35. Divide both sides by __. Then x = __.

D. Omar solves 6x = 42 by writing x = 42 − 6 = 36. Explain and repair his work.

M1. 24 shells are 30% of a collection. Find the whole collection.

M2. Find 2 ÷ $\frac{2}{3}$.

M3. Find 3.45 + 5.078.

X1. Solve x + 4 = 7. Show a change to both sides.

S. I can check a solution by __.

E. Solve $(\frac{3}{5})$ x = 12. Show equal parts or an equation step. Check your value.

CHALLENGE / Explain your plan and check the result

C1. Three identical snack bags and a $4 delivery fee cost $19. Find the price per bag. A second shop charges $6 per bag with no fee. Which shop is cheaper for five bags, and by how much?

C2. A tank starts with 3 liters and gains 2 liters each minute. Make a table for 0, 1, 3, 5 minutes. Write a rule. When will it hold 17 liters? Explain why multiplying the time by 2 alone is insufficient.

Week 6 reference

Expressions name quantities. Equivalent forms describe the same quantity. Equations can be solved by keeping both sides equal.

Expressions and equivalent forms

An expression names a quantity. Equivalent expressions name the same quantity for every allowed value.

Example: Find $3(2 + 4)$.

The parentheses name one group: $2 + 4 = 6$.

Three groups give $3 \times 6 = 18$.

Split each group instead: $3 \times 2 + 3 \times 4 = 6 + 12 = 18$.

Result: 18

Equations and related quantities

A solution makes the equation true. The same operation on both sides preserves equality.

Example: Solve $x + 2.75 = 8$.

Both sides must stay equal.

Subtract 2.75 from each side: $x = 5.25$.

Check: $5.25 + 2.75 = 8$.

Result: $x = 5.25$

Words to use

variable: A letter that can stand for a number.

expression: Numbers, variables, and operations without an equals sign.

exponent: The number of equal factors to multiply.

equivalent: Having the same value for every allowed input.

equation: A statement that two quantities are equal.

Expressions and equations

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 8×3

2. 8×7

3. 8×4

4. 8×9

5. A rectangle has area $\frac{4}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

6. Find $\frac{2}{3} + \frac{1}{4}$.

7. 2×4

8. Find $2n + 7$ when $n = 5$.

9. Expand $2(x + 7)$.

10. Find 5^2. Write the repeated factors.

11. Find $4n + 3$ when $n = 2$.

12. Each pack has b beads. Write the total in 3 packs. Find it when $b = 4$.

13. Lena wrote $5(z + 2) = 5z + 2$. Find the mistake and repair it.

14. A packet contains 5 cards. Write the number in p packets plus 4 loose cards; evaluate at $p = 4$.

15. Evaluate $4t + 5$ when $t = 4$. Show the multiplication first.

16. Evaluate $4^2 + 5$. Explain what the exponent means.

17. Expand $4(x + 5)$ and check at $x = 4$.

18. Simplify $5y + 4y + y$. Evaluate at $y = 2$.

CHALLENGE / Explain your plan and check the result

C1. A museum sells family packs with 2 adult tickets and 3 child tickets. Adult tickets cost a dollars; child tickets cost $4. Write the cost of 4 packs. If $a = $9, will $125 cover the order? Explain.

C2. A rectangular garden is 5 m wide and $n + 2$ m long. A path covers 10 square m inside it. Write the planted area. Find it when $n = 6$, and explain why $5n$ is equivalent to your expression.

Expressions and equations

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find $\frac{4}{7} \div \frac{2}{5}$. Check the result by multiplying by $\frac{2}{5}$.

6. 12 shells are 30% of a collection. Find the whole collection.

7. $36 \div 3$

8. Solve $(\frac{2}{5})\ x = 10$.

9. Solve $x + 11 = 18$.

10. Solve $7x = 56$.

11. Solve $1.5x = 4.80$.

12. Solve $(\frac{3}{4})\ x = 12$.

13. Omar solves $6x = 42$ by writing $x = 42 - 6 = 36$. Explain and repair his work.

14. A packet contains 6 cards. Write the number in p packets plus 5 loose cards; evaluate at $p = 5$.

15. Evaluate $5t + 6$ when $t = 5$. Show the multiplication first.

16. Evaluate $5^2 + 6$. Explain what the exponent means.

17. Expand $5(x + 6)$ and check at $x = 5$.

18. Simplify $6y + 5y + y$. Evaluate at $y = 2$.

CHALLENGE / Explain your plan and check the result

C1. A ribbon is cut into four equal pieces. After 1.5 m is trimmed from each piece, each remaining piece is 2.25 m. Find the original ribbon length. Write an equation and check it.

C2. Five friends split a taxi bill equally. The fare is $18 plus $2 per kilometer. Each friend pays $6. Find the trip length. Then find each person’s share if the same trip had six riders.

Expressions and equations

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 8.2 × 10

2. 800 ÷ 100

3. Double 8.5

4. Half of 48

5. Solve 1.5x = 4.80.

6. Expand 4(x + 5).

7. Find 0.25 × 0.02.

8. Find 288 ÷ 24.

9. 12 shells are 30% of a collection. Find the whole collection.

10. Red : blue is 3 : 5. How many blue tiles go with 9 red tiles?

11. 4 × 6

12. A box is 4 cm by 6 cm by 3 cm. Find its volume.

13. Two whole numbers total 17. One is 5 more than the other. Find both numbers and show a way to check.

14. A packet contains 7 cards. Write the number in p packets plus 2 loose cards; evaluate at p = 6.

15. Evaluate 2t + 7 when t = 6. Show the multiplication first.

16. Evaluate $2^2 + 7$. Explain what the exponent means.

17. Expand 2(x + 7) and check at x = 6.

18. Simplify 7y + 2y + y. Evaluate at y = 2.

19. A rectangle has height 2 and width z + 7. Write its area two ways.

20. Solve x + 7 = 13. Show a change to both sides.

21. Solve 2x = 12. Check using multiplication.

22. Solve $\frac{2}{3}$ x = 12. Draw 3 equal parts.

23. 7 identical pens cost $14. Write and solve an equation for price p per pen.

CHALLENGE / Explain your plan and check the result

C1. A club buys 8 notebooks at n dollars each and returns 2. It also buys pens for $9. Its final bill is $33. Write two equivalent bill expressions, find n, and check the price against the original purchase and return.

C2. A number machine multiplies by 3, then adds 5. A second machine adds 5, then multiplies by 3. Compare outputs for input 4. How much greater is the second output for any input? Explain with expressions.

Expressions and equations

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 848 to the nearest hundred.

2. Round 8.72 to the nearest whole number.

3. Use 20×5 to estimate 19.8×5.1.

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

5. Solve $1.5x = 4.80$.

6. Expand $4(x + 5)$.

7. Solve $7x = 56$.

8. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

9. Find $9.6 \div 8$.

10. Find $2.8 \div 0.07$.

11. A ball costs \$4.65 and a card costs \$2.78. Find the change from \$10.

12. Find $10 - 1.286$.

13. Each pack has b beads. Write the total in 6 packs. Find it when $b = 4$.

14. Omar solves $6x = 42$ by writing $x = 42 - 6 = 36$. Explain and repair his work.

15. A packet contains 8 cards. Write the number in p packets plus 3 loose cards; evaluate at $p = 7$.

16. Evaluate $3t + 8$ when $t = 7$. Show the multiplication first.

17. Evaluate $3^2 + 8$. Explain what the exponent means.

18. Expand $3(x + 8)$ and check at $x = 7$.

CHALLENGE / Explain your plan and check the result

C1. A shop has a \$45.75 budget for identical folders and a \$9 delivery charge. Seven folders fit the budget exactly. Find the folder price. Can the shop buy 9 folders if delivery becomes free? Show the remaining amount.

C2. A square tile has side x cm. A border strip adds a rectangle 2 cm wide along one side. Write the total area. Compare $x = 1$, $x = 3$, and $x = 5$. Do the two increases of 2 in x produce equal increases in area?

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Solve 1.5x = 4.80.

2. Expand 4(x + 5).

3. Solve 7x = 56.

4. Find 3n + 1 when n = 4.

5. Find 10 − 1.286.

6. Find 0.25 × 0.04.

7. Find 0.25 × 0.02.

8. 12 shells are 30% of a collection. Find the whole collection.

9. Red : blue is 3 : 5. How many blue tiles go with 9 red tiles?

10. Lena wrote 5(z + 2) = 5z + 2. Find the mistake and repair it.

11. A packet contains 9 cards. Write the number in p packets plus 4 loose cards; evaluate at p = 8.

12. Evaluate 4t + 9 when t = 8. Show the multiplication first.

13. Evaluate 4^2 + 9. Explain what the exponent means.

14. Expand 4(x + 9) and check at x = 8.

15. Simplify 9y + 4y + y. Evaluate at y = 2.

16. A rectangle has height 4 and width z + 9. Write its area two ways.

17. Solve x + 9 = 17. Show a change to both sides.

18. Solve 4x = 32. Check using multiplication.

CHALLENGE / Explain your plan and check the result

C1. A craft club chooses between A: $12 setup plus $3 per bracelet and B: $5 per bracelet. Find both costs for 4 and 8 bracelets. Find a number of bracelets for which the costs match and explain your reasoning.

C2. A student claims 2(x + 5) + 3x equals 7x for every x because 2 + 5 = 7. Test x = 2, repair the expression, and find whether the two expressions happen to agree for any positive whole-number input.

Area from a base and height

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Find $2\frac{1}{2} \div \frac{3}{4}$.

R2. Three notebooks cost $2.45 each. What is the total cost?

R3. Find 3^2. Write the repeated factors.

R4. Find $4 \div \frac{2}{3}$.

P1. A rectangle is 6 cm long and 2.5 cm wide. Find its area and perimeter.

P2. A parallelogram has base 7 cm, height 4 cm, and slanted side 5 cm. Cut and move the end triangle. Find its area.

P3. A right triangle has perpendicular legs of 6 cm and 8 cm. Find its area. Use either leg as the base.

P4. An isosceles triangle has sides 13, 13, and 10 cm. Its height to the 10 cm base is 12 cm. Find the area.

P5. A triangle has vertices A(0, 0), B(4, 0), and C(6, 3), measured in cm. The height meets the extended base at (6, 0). Find its area.

P6. A triangle has base 5 cm and matching height 2.4 cm. Find its area. Would base × slanted side always work?

F. A triangle has base 8 cm and height 2.5 cm. Complete: $A = \frac{1}{2} \times 8 \times __ = __$.

D. Theo uses a triangle's base 9 cm and height 4 cm. He writes $9 \times 4 = 36$ square cm. What is missing?

X1. A rectangle has length 6 cm and width 3 cm. Find area and perimeter.

X2. A parallelogram has base 4 cm, height 3 cm, and slanted side 5 cm. Find area and name the unused length.

X3. A right triangle has perpendicular legs 8 cm and 3 cm. Find area using either leg as base.

X4. A triangle has area 12 square cm and base 8 cm. Find its matching height.

S. For this base, the matching height is __ because __.

E. A triangle has sides 10, 10, and 12 cm. The height to the 12 cm base is 8 cm. Find its area and name the length you used as height.

CHALLENGE / Explain your plan and check the result

C1. A parallelogram garden has base 12 m, perpendicular height 5 m, and slanted side 7 m. A triangular pond with base 6 m and height 4 m lies inside it. Find the planted area. Explain why using 7 m as the garden height changes the result incorrectly.

C2. A triangle has base 10 cm and height 12 cm. Design a rectangle of the same area with one side 5 cm. Compare their areas, then explain whether equal area guarantees equal perimeter. Use the triangle sides 10, 13, 13 cm for the perimeter comparison.

Nets and surface area

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

R2. Find $10 - 2.286$.

R3. Solve $x + 9 = 23$.

R4. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

P1. An 8 cm by 6 cm rectangle loses a 3 cm by 2 cm corner from the upper right. Find the L-shaped area two ways.

P2. A closed box is 2 cm by 3 cm by 4 cm. Label all six rectangles on a net. Find surface area and volume.

P3. A triangular prism has right-triangle ends with sides 6, 8, and 10 cm. Its length is 5 cm. List the five faces and find surface area.

P4. A square pyramid has base side 6 cm. Each triangular face has perpendicular height 5 cm. Find its surface area.

P5. A closed 2 cm by 4 cm by 5 cm box has a claimed surface area of 68 square cm. Its 2 by 4 bottom was left out. Repair the answer.

F. A square pyramid has base side 4 cm and face height 3 cm. Complete: SA = 16 + 4 × ($\frac{1}{2}$ × 4 × __) = __.

D. Ava finds a cube's surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

M1. A ribbon is $4\frac{1}{2}$ meters long. Pieces measure $\frac{3}{8}$ meter. How many pieces fit?

M2. Find $312 \div 24$.

M3. Use $y = 2x$ for $x = 1, 3, 5$. Give the ordered pairs.

X1. A closed box measures 2 cm by 4 cm by 5 cm. Find surface area.

X2. A cube has side 4 cm. Find surface area and volume; label units.

S. To find surface area, I list __ and add __.

E. A triangular prism has right-triangle ends with sides 3, 4, and 5 cm. It is 8 cm long. Find the area of all five faces together.

CHALLENGE / Explain your plan and check the result

C1. A closed rectangular box is 4 cm by 3 cm by 2 cm. Find its surface area. A second version has no 4-by 3 top and needs an extra 2-by 3 label made from the same material. How much material does the second version need, and how much is saved?

C2. A triangular prism has right-triangle ends with sides 3, 4, 5 cm and length 10 cm. Find surface area and volume. A builder claims that doubling the length doubles both results. Test the claim by using length 20 cm.

Week 7 reference

Area counts square units that cover a surface. Cutting and rearranging reveal triangle and parallelogram formulas. A net helps count every face of a solid.

Area from a base and height

A parallelogram rearranges into a rectangle. Two copies of a triangle make a parallelogram.

Example: A parallelogram has base 6 cm and height 3 cm.

Cut a triangle from one end and move it to the other.

The new rectangle has the same area, base 6 and height 3.

Area = 6 × 3 = 18 square cm.

Result: 18 square cm

Nets and surface area

A net unfolds every face. Surface area adds face areas in square units.

Example: Find the surface area of a 2 cm by 2 cm by 3 cm closed box.

Two square ends each have area 4.

Four side rectangles each have area 6.

Total area = 2 × 4 + 4 × 6 = 32 square cm.

Result: 32 square cm

Words to use

area: The amount of flat surface inside a shape.

base: The side chosen to match a height.

height: The perpendicular distance from the base line to the opposite side or vertex.

composite polygon: A shape made by joining simpler polygons.

net: A flat arrangement that folds into a solid.

Area and surface area

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 9×3

2. 9×7

3. 9×4

4. 9×9

5. Find $1{,}344 \div 32$. Verify by multiplication.

6. 9 of 12 tiles are green. What percent are green?

7. 3×5

8. A parallelogram has base 6, height 8, and slanted side 10 cm. Find its area.

9. A rectangle is 6 cm by $\frac{3}{2}$ cm. Find its area.

10. A parallelogram has base 9, height 4, and slanted side 5 cm. Find its area.

11. A triangle has base 9 cm and height 4 cm. Find its area.

12. A 9 by 5 cm rectangle loses a 2 by 3 cm corner. Find the remaining area.

13. Theo uses a triangle’s base 9 cm and height 4 cm. He writes $9 \times 4 = 36$ square cm. What is missing?

14. A rectangle has length 8 cm and width 5 cm. Find area and perimeter.

15. A parallelogram has base 6 cm, height 5 cm, and slanted side 7 cm. Find area and name the unused length.

16. A right triangle has perpendicular legs 12 cm and 5 cm. Find area using either leg as base.

17. A triangle has area 30 square cm and base 12 cm. Find its matching height.

18. A9 cm by 7 cm rectangle loses a3 cm by 2 cm corner. Find remaining area.

CHALLENGE / Explain your plan and check the result

C1. A wall is 8 m wide and 3 m high. It contains a2 m by 1.5 m window and a triangular vent with base 2 m and height 1 m. Paint covers 5 square m per liter. How many whole liters must be bought?

C2. Two triangular flags each have area 24 square cm. Flag A has base 8 cm; flag B has base 6 cm. Find both heights. Which flag is taller, and by what factor? Explain using area, not a sketch’s appearance.

Area and surface area

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find 3^2. Write the repeated factors.

6. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

7. $48 \div 4$

8. Find the surface area of a closed 3 by 3 by 4 cm box.

9. Find the surface area of a closed 2 by 3 by 5 cm box.

10. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

11. A square pyramid has base side 4 cm and triangular-face height 3 cm. Find surface area.

12. A closed cube has side 2 cm. Jo counts five faces and gets 20 square cm. Repair the total.

13. Ava finds a cube's surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

14. A rectangle has length 9 cm and width 6 cm. Find area and perimeter.

15. A parallelogram has base 7 cm, height 6 cm, and slanted side 8 cm. Find area and name the unused length.

16. A right triangle has perpendicular legs 14 cm and 6 cm. Find area using either leg as base.

17. A triangle has area 42 square cm and base 14 cm. Find its matching height.

18. A10 cm by 8 cm rectangle loses a3 cm by 2 cm corner. Find remaining area.

CHALLENGE / Explain your plan and check the result

C1. Two closed boxes have dimensions 2 × 3 × 6 cm and 3 × 3 × 4 cm. Find each volume and surface area. Which uses less cardboard for the same capacity, and by how much?

C2. A square pyramid has base side 8 cm and triangular face height 5 cm. Only its four triangular faces are painted. Paint costs $0.15 per square cm. Find the paint cost; then find the extra cost if the base is painted too.

Area and surface area

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 9.2 × 10

2. 900 ÷ 100

3. Double 9.5

4. Half of 54

5. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

6. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

7. Solve x + 9 = 23.

8. Find 0.8 × 0.7. Is the product below both factors?

9. Find $\frac{3}{4} \times \frac{4}{5}$.

10. Order $\frac{2}{5}$, 0.45, and $\frac{1}{2}$ from least to greatest.

11. 5 × 2

12. A box is 5 cm by 2 cm by 3 cm. Find its volume.

13. A rectangle has whole-number side lengths, perimeter 24 cm, and area 35 square cm. Find its side lengths.

14. A rectangle has length 10 cm and width 3 cm. Find area and perimeter.

15. A parallelogram has base 8 cm, height 3 cm, and slanted side 5 cm. Find area and name the unused length.

16. A right triangle has perpendicular legs 16 cm and 3 cm. Find area using either leg as base.

17. A triangle has area 24 square cm and base 16 cm. Find its matching height.

18. A11 cm by 5 cm rectangle loses a3 cm by 2 cm corner. Find remaining area.

19. A triangle and parallelogram share base 8 and height 3. Find each area and their ratio.

20. A closed box measures 2 cm by 3 cm by 8 cm. Find surface area.

21. A cube has side 3 cm. Find surface area and volume; label units.

22. A square pyramid has base side 6 cm and triangular face height 8 cm. Find surface area.

23. A prism has 3-4-5 right-triangle ends and length 8 cm. Find its surface area.

CHALLENGE / Explain your plan and check the result

C1. An L-shaped floor comes from a10 m by 8 m rectangle with a4 m by 3 m corner removed. Tiles cover 2 square m each. Tiles are sold in packs of 6. How many packs must be bought, and how much area will the unused tiles cover?

C2. A tent is a triangular prism 8 m long with right-triangle ends of sides 3, 4, 5 m. The 3 m side rests on the ground. Find the fabric area for both ends and the two other rectangular sides, with no floor. Compare with a fully closed tent.

Area and surface area

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 948 to the nearest hundred.

2. Round 9.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

6. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

7. Find the surface area of a closed 2 by 3 by 5 cm box.

8. Each pack has b beads. Write the total in 8 packs. Find it when b = 4.

9. Expand 3(x + 6).

10. Find 2□. Write the repeated factors.

11. Solve 6x = 54.

12. Use y = 4x for x = 0, 2, 3. Give the ordered pairs.

13. A prism has right-triangle ends with sides 3, 4, 5 cm and length 9 cm. Find surface area.

14. Ava finds a cube’s surface area by adding five faces. Each face is 9 square cm. She gets 45. Explain and repair it.

15. A rectangle has length 11 cm and width 4 cm. Find area and perimeter.

16. A parallelogram has base 9 cm, height 4 cm, and slanted side 6 cm. Find area and name the unused length.

17. A right triangle has perpendicular legs 18 cm and 4 cm. Find area using either leg as base.

18. A triangle has area 36 square cm and base 18 cm. Find its matching height.

CHALLENGE / Explain your plan and check the result

C1. A rectangular sign is 9 cm by 7 cm. A triangle of base 6 cm and perpendicular height 4 cm is cut away. The remaining sign is copied twice more. Find total remaining area for all three copies. Explain a way to calculate without repeating every subtraction.

C2. A cube has side 4 cm. One student counts only the four side faces; another counts all six. Find both totals and the difference. The cube is then opened by removing its top. Which total describes the open box?

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

2. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

3. Find the surface area of a closed 2 by 3 by 5 cm box.

4. A square pyramid has base side 6 cm and triangular-face height 5 cm. Find surface area.

5. Find $4n + 3$ when $n = 2$.

6. Find $3 \div 8$.

7. Solve $1.5x = 4.80$.

8. Find $\frac{3}{4} \times \frac{4}{5}$.

9. Order $\frac{2}{5}$, 0.45, and $\frac{1}{2}$ from least to greatest.

10. Theo uses a triangle's base 9 cm and height 4 cm. He writes $9 \times 4 = 36$ square cm. What is missing?

11. A rectangle has length 12 cm and width 5 cm. Find area and perimeter.

12. A parallelogram has base 10 cm, height 5 cm, and slanted side 7 cm. Find area and name the unused length.

13. A right triangle has perpendicular legs 20 cm and 5 cm. Find area using either leg as base.

14. A triangle has area 50 square cm and base 20 cm. Find its matching height.

15. A13 cm by 7 cm rectangle loses a3 cm by 2 cm corner. Find remaining area.

16. A triangle and parallelogram share base 10 and height 5. Find each area and their ratio.

17. A closed box measures 2 cm by 5 cm by 10 cm. Find surface area.

18. A cube has side 5 cm. Find surface area and volume; label units.

CHALLENGE / Explain your plan and check the result

C1. A parallelogram has area 72 square cm and base 9 cm. A triangle has the same base and area. Find their perpendicular heights. Explain why the triangle requires twice the height, then find its area if its height is reduced by 3 cm.

C2. A closed $4 \times 5 \times 6$ cm box is wrapped with paper. A label covers 20 square cm already, so no new paper is needed there. Paper costs \$0.02 per square cm. Find the paper area and cost. Explain why volume is not the right quantity.

Negative numbers and distance from zero

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Find 3.45 + 5.078.

R2. Solve x + 11 = 18.

R3. A parallelogram has base 6, height 8, and slanted side 10 cm. Find its area.

R4. Three notebooks cost $2.45 each. What is the total cost?

P1. Write a signed number for 4 degrees below zero, 7 m above sea level, and owing $9. State what zero means in each setting.

P2. Place −6, 1, and −2 on a number line. Which is least?

P3. Place −2.5, $-\frac{3}{4}$, and 0.5 on a number line. Order them from least to greatest.

P4. Find the opposite of −9. Then find the opposite of that result. Write −(−9).

P5. Find |−8|, |8|, and |0|. Describe each as a distance.

P6. Compare −9 and −4 using < or >. Compare their absolute values. Which temperature is warmer, −9°C or −4°C?

F. Complete: −7 is __ units from zero, so |−7| = __. Its opposite is __.

D. Ben says −10 > −3 because 10 > 3. Use a number line to explain and repair his statement.

X1. Write signed numbers for 3 meters below sea level and 2 meters above it. State what zero represents.

X2. Order these values from least to greatest: $\frac{3}{2}$, -3, 0, -1.

X3. Find the opposite of − 3, then its opposite. Explain why you return to the start.

X4. Find|−3|, |3|, and|0|. Compare the first two.

S. The greater number lies __. The greater absolute value means __.

E. Order −6, 1.5, $-\frac{1}{4}$, and −2.5. Which number has the greatest absolute value? Give that absolute value.

CHALLENGE / Explain your plan and check the result

C1. Three storage rooms are at − 6°C, −1°C, and 4°C. Order them from coldest to warmest. Which is closest to 0°C? Find the temperature gap between the coldest and warmest rooms, and explain why the largest absolute value does not identify the warmest room.

C2. A diver is 12 m below sea level; a platform is 5 m above sea level. Write both signed locations. Find the distance from the diver to sea level and to the platform. Is the platform at the opposite of the diver’s location? Explain.

Coordinates and inequalities

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Find 1.4 ÷ 0.07.

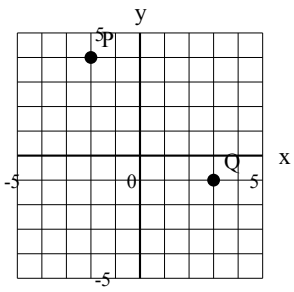
R2. Solve $(\frac{2}{5})$ x = 10.

R3. Find the surface area of a closed 2 by 3 by 5 cm box.

R4. Find 10 − 2.286.

P1. Plot A(2, 3), B(−4, 1), C(−3, −2), and D(1, −4). Name the quadrant of each.

P2. Read P and Q from the grid. State the horizontal move first.



P3. Reflect (−2, 3) across the x-axis. Then reflect the original point across the y-axis.

P4. Find the distance from (−2, 3) to (−2, −4), and from (−2, 3) to (5, 3).

P5. A rectangle has vertices (−3, 2), (2, 2), (2, −1), and (−3, −1). Find its perimeter.

P6. Graph x > 2 and x ≥ 2 on separate number lines. Is 2 a solution of each?

P7. A ride requires a height of at least 48 inches. Write an inequality for h. Is 48 included?

F. Reflect (−5, −2) across the y-axis: (__, __). The distance between the points is __.

D. Nia writes x > 6 for “at least 6.” She leaves 6 out. Explain and repair her graph.

M1. Find 1.45 + 3.078.

M2. Find 3n + 1 when n = 4.

M3. A rectangle is 8 cm by 2 cm. Find its area.

S. A reflection across the y-axis changes __ and keeps __.

E. Reflect A(−4, 1.5) across the y-axis. Give the image and the distance from A to its image.

CHALLENGE / Explain your plan and check the result

C1. A delivery route goes from A(−4, 2) to B(3, 2), then to C(3, −3). Find the route length. Reflect all three points across the y-axis and give the new route length. Explain why reflection preserves the distance.

C2. A club accepts whole-number ages of at least 10 but less than 14. A second activity is for ages at most 12. List the ages allowed in both. Write each inequality and explain whether ages 10, 12, and 14 are included.

Week 8 reference

A signed number gives a position relative to zero. Coordinates use two such positions. Inequalities describe all positions that meet a condition.

Negative numbers and distance from zero

Numbers increase to the right. Absolute value measures distance from zero, regardless of direction.

Example: Compare −1.5 and −0.5.

Locate zero, then mark both numbers to its left.

−1.5 lies one unit left of −0.5.

Therefore $-1.5 < -0.5$, but $|-1.5| > |-0.5|$.

Result: $-1.5 < -0.5$

Coordinates and inequalities

A coordinate pair gives horizontal then vertical position. An inequality can describe many solutions.

Example: Find the distance from (−1, −3) to (−1, 2).

The x-coordinate stays −1, so the segment is vertical.

From −3 to 0 is 3 units. From 0 to 2 is 2 units.

Total distance = 3 + 2 = 5 units.

Result: 5 units

Words to use

negative number: A number less than zero.

opposites: Numbers the same distance from zero on opposite sides.

absolute value: A number’s distance from zero.

coordinate pair: Two numbers naming horizontal and vertical position.

reflection: A mirror image across a line.

Negative numbers, coordinates, and inequalities

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 10×3

2. 10×7

3. 10×4

4. 10×9

5. A parallelogram has base 6, height 8, and slanted side 10 cm. Find its area.

6. 9 of 12 tiles are green. What percent are green?

7. 4×6

8. Find $|5|$ and describe it as a distance.

9. Order -4, 0, -9 from least to greatest.

10. Find $|-12|$ and describe it as a distance.

11. Find $|0|$ and describe it as a distance.

12. Find the opposite of -6.

13. Ben says $-10 > -3$ because $10 > 3$. Use a number line to explain and repair his statement.

14. Write signed numbers for 5 meters below sea level and 4 meters above it. State what zero represents.

15. Order these values from least to greatest: $\frac{3}{2}$, -5, 0, -2.

16. Find the opposite of -5, then its opposite. Explain why you return to the start.

17. Find $|-5|$, $|5|$, and $|0|$. Compare the first two.

18. Which is warmer, $-5°C$ or $-7°C$? Which has greater absolute value?

CHALLENGE / Explain your plan and check the result

C1. A trail marker is at -7 km on an east-west line. Another is its opposite. Find the distance between them. A shelter at -2 km lies between them; find both distances from the shelter and check that they add to the whole distance.

C2. Find a negative number that is 10 units from 3. Then find its opposite. Compare the two numbers and their absolute values. Explain why the larger number does not have the larger absolute value.

Negative numbers, coordinates, and inequalities

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find the surface area of a closed 2 by 3 by 5 cm box.

6. Five granola bars are shared equally among six children. How much does each get?

7. 60 ÷ 5

8. Find the distance from (1, -5) to (1, 4).

9. Reflect (-3, 4) across the y-axis.

10. Find the distance from (-4, 2) to (3, 2).

11. Write an inequality for x that means at least 9. Is the boundary included?

12. Write an inequality for x that means less than 5. Is the boundary included?

13. Nia writes x > 6 for “at least 6.” She leaves 6 out. Explain and repair her graph.

14. Write signed numbers for 6 meters below sea level and 5 meters above it. State what zero represents.

15. Order these values from least to greatest: $\frac{3}{2}$, -6, 0, $\frac{-5}{2}$.

16. Find the opposite of − 6, then its opposite. Explain why you return to the start.

17. Find|−6|, |6|, and|0|. Compare the first two.

18. Which is warmer, −6°C or − 8°C? Which has greater absolute value?

CHALLENGE / Explain your plan and check the result

C1. A rectangle has vertices(−5, −2), (1, −2), (1, 3), (−5, 3). Find its side lengths, perimeter, and area. Reflect it across the x-axis. Give the reflected point corresponding to(−5, 3) and explain which measurements change.

C2. A package may weigh at most 8 kg. Its empty box weighs 1.5 kg. Two identical items are packed inside. Write an inequality for the weight w of each item. Test w = 3, w = 3.25, andw = 3.5 kg.

Negative numbers, coordinates, and inequalities

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 10.2 × 10

2. 1000 ÷ 100

3. Double 10.5

4. Half of 60

5. Find |0| and describe it as a distance.

6. Find the opposite of 9.

7. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

8. A 10 by 4 cm rectangle loses a 3 by 1 cm corner. Find the remaining area.

9. Is $\frac{4}{3}$ × 6 smaller or larger than 6? Compute to check.

10. Find 25% of 24 stickers.

11. List common factors of 12 and 18. Find the greatest common factor.

12. List common factors of 8 and 20. Find the greatest common factor.

13. A point lies left of the y-axis and above the x-axis. It is 3 units from the y-axis and 2 from the x-axis. Find it, then reflect it across both axes in turn.

14. Write signed numbers for 7 meters below sea level and 2 meters above it. State what zero represents.

15. Order these values from least to greatest: $\frac{3}{2}$, -7, 0, -1.

16. Find the opposite of − 7, then its opposite. Explain why you return to the start.

17. Find|−7|, |7|, and|0|. Compare the first two.

18. Which is warmer, −7°C or − 9°C? Which has greater absolute value?

19. Find the distance between − 7 and 2 on a number line. Split the distance at zero.

20. Name the quadrant of each point: A(2, 7), B(−2, 7), C(−2, −7), D(2, −7).

21. Reflect(−2, 7) across the x-axis and then, starting again, across the y-axis.

22. Find the distance from(−2, −2) to(7, −2). Explain why only x changes.

23. A rectangle has vertices(−2, 0), (7, 0), (7, 3), (−2, 3). Find perimeter.

CHALLENGE / Explain your plan and check the result

C1. Points A and B lie on a horizontal line at y = −2. A is(−3, −2), and B is 8 units from A. Give both possible B points. Which is in quadrant IV? Reflect that point across the y-axis.

C2. Two shuttle lights flash every 12 seconds and 18 seconds. They flash together at time 0. Find the next shared flash and the number of flashes each light makes after time 0 up to and including that moment. Explain why GCF is not the time you need.

Negative numbers, coordinates, and inequalities

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 1048 to the nearest hundred.

2. Round 10.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. Find |0| and describe it as a distance.

6. Find the opposite of 9.

7. Reflect (-3, 4) across the y-axis.

8. A triangle has base 9 cm and height 4 cm. Find its area.

9. A rectangle is 7 cm by 3 cm. Find its area.

10. A 7 by 6 cm rectangle loses a 2 by 2 cm corner. Find the remaining area.

11. Find the surface area of a closed 3 by 3 by 4 cm box.

12. A box has edges $\frac{1}{2}$, $\frac{2}{3}$, and 2 meters. Find the volume.

13. Write an inequality for x that means at least 9. Is the boundary included?

14. Nia writes x > 6 for “at least 6.” She leaves 6 out. Explain and repair her graph.

15. Start at −3. Move 5 units right. Where do you land?

16. Solve 2x + 3 = 11.

17. Write signed numbers for 8 meters below sea level and 3 meters above it. State what zero represents.

18. Order these values from least to greatest: $\frac{3}{2}$, -8, 0, $\frac{-3}{2}$.

CHALLENGE / Explain your plan and check the result

C1. A safe freezer temperature t must be at least − 8°C but below − 3°C. Test − 9, −8, −5, −3, and 0. Graph the allowed interval and explain why − 8 is included while − 3 is not.

C2. A bird starts at(−2, 4), flies straight down 7 units, then straight right 6 units. Give the two new positions and total distance. Another bird flies right 6 then down 7 from the same start. Does it finish at the same point? Explain.

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Find |0| and describe it as a distance.

2. Find the opposite of 9.

3. Reflect (-3, 4) across the y-axis.

4. Find |−12| and describe it as a distance.

5. Solve 6x = 54.

6. Solve 1.5x = 4.80.

7. A square pyramid has base side 4 cm and triangular-face height 3 cm. Find surface area.

8. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

9. Find 25% of 24 stickers.

10. Ben says −10 > −3 because 10 > 3. Use a number line to explain and repair his statement.

11. Write signed numbers for 9 meters below sea level and 4 meters above it. State what zero represents.

12. Order these values from least to greatest: $\frac{3}{2}$, -9, 0, -2.

13. Find the opposite of − 9, then its opposite. Explain why you return to the start.

14. Find|−9|, |9|, and|0|. Compare the first two.

15. Which is warmer, −9°C or − 11°C? Which has greater absolute value?

16. Find the distance between − 9 and 4 on a number line. Split the distance at zero.

17. Name the quadrant of each point: A(4, 9), B(−4, 9), C(−4, −9), D(4, −9).

18. Reflect(−4, 9) across the x-axis and then, starting again, across the y-axis.

CHALLENGE / Explain your plan and check the result

C1. A number-line game allows whole numbers x with −4≤x≤3. A bonus requires|x|≥3. List all bonus numbers and their opposites. Which opposites remain allowed in the original game?

C2. A square has consecutive vertices(−2, −2), (3, −2), (3, 3), (−2, 3). Find its perimeter. Reflect the square across the y-axis. What is the leftmost x-coordinate of the image? Explain why a vertex originally left of the axis can move right of it.

Data displays and center

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. Use y = 2x for x = 1, 3, 5. Give the ordered pairs.

R2. Find the surface area of a closed 2 by 4 by 6 cm box.

R3. Find the distance from (-2, -3) to (-2, 2).

R4. Solve x + 11 = 18.

P1. Which asks a statistical question: “How many pages are in this book?” or “How many pages are in each library book on this shelf?” Explain.

P2. Build a dot plot for 1, 2, 2, 3, 4, 4, 5 books read. Describe its shape and total number of dots.

P3. Build a histogram for 0, 1, 1, 2, 2, 3, 4, 5. Use bins 0 to under 2, 2 to under 4, and 4 to under 6.

P4. Four jars hold 2, 4, 5, and 9 counters. Share all counters equally. Find the mean and show how it balances the deviations.

P5. Find each median: (a) 8, 2, 4, 4, 7; (b) 5, 2, 9, 3, 6, 5. Order first.

P6. Data are 2, 3, 4, 4, 7. Replace 7 with 27. Compare the old and new mean and median. Which center changes less?

P7. Four game scores have mean 6. Three scores are 3, 5, and 7. Find the missing score.

F. Ordered data are 1, 3, 6, 8. Complete: median = (3 + __) ÷ 2 = __.

D. Luis finds the median of 9, 2, 6, 3 by averaging 2 and 6. He gets 4. What should he change?

X1. Choose the statistical question: (A) How many pages are in one named book?(B) How many pages are in each of 8 books? Explain.

X2. Make a dot plot for 1, 1, 2, 3, 3, 3, 5. State the total dots and most frequent value.

X3. Find histogram frequencies for 0, 1, 2, 2, 3, 4, 5, 5 using bins 0 to under 2, 2 to under 4, and 4 to under 6.

S. The mean tells __. The median tells __.

E. Data are 8, 2, 11, 5, 6, 4. Find the median and the mean. Show your ordered list.

CHALLENGE / Explain your plan and check the result

C1. Seven reading times are 8, 9, 10, 10, 11, 12, and 24 minutes. Find the mean and median. Replace 24 with 10 and recompute. Which center changes more, and which original center better describes most of the times? Support your choice.

C2. Five quiz scores have mean 14. Four scores are 10, 12, 16, and 18. Find the missing score and median. Then explain whether adding a sixth score of 14 changes the mean or the median.

Spread and box plots

20 questions, including 2 challenges. Write your work and answers on separate paper.

P1. Data are 3, 7, 8, 10, 12. Find the range.

P2. Find the five-number summary of 1, 3, 4, 6, 8, 9, 11. Exclude the median from each half.

P3. Find the five-number summary and IQR of 1, 2, 4, 6, 8, 10, 12, 13.

P4. Data are 2, 4, 6, 8. Find the mean, each distance from the mean, and the MAD.

P5. A box plot has five-number summary 0, 2.5, 5, 9, 12. Which interval represents the upper quarter? Does a longer interval mean more data?

P6. Compare A: 4, 4, 6, 6 and B: 1, 3, 7, 9. Find each mean and MAD. Which set is more spread out?

P7. Data are 1, 3, 4, 5, 6, 7, 9, 11. Replace 11 with 31. Compare the old and new range and IQR.

F. Data are 3, 5, 7, 9. Mean = 6. Distances are 3, __, __, 3. MAD = __.

D. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

X1. Find the range of 2, 4, 6, 9, 12. Which values determine it?

X2. Find the five-number summary and IQR of 2, 3, 5, 7, 9, 11, 13. Exclude the median from each half.

X3. Find the five-number summary and IQR of 2, 4, 5, 7, 9, 11, 12, 14.

X4. Find the mean and MAD of 2, 4, 8, 10. Show all four distances.

X5. A box plot has summary(1, 3, 6, 10, 13). Find range and IQR, and name the interval containing the upper quarter.

X6. Compare MAD for A: 5, 5, 7, 7 and B: 2, 4, 8, 10. Both means are 6.

X7. Data are 2, 3, 5, 7, 9, 11, 13, 14. Replace 14 with 34. Compare the old and new range and IQR.

S. IQR measures __. MAD measures __.

E. For 1, 3, 3, 5, 7, 9, 11, 13, find Q1, the median, Q3, and the IQR. Show the two halves.

CHALLENGE / Explain your plan and check the result

C1. Compare A: 2, 4, 6, 8 and B: 0, 4, 6, 10. Find each mean, range, and MAD. Which has the same center but greater spread? Describe what changed in the data.

C2. For 2, 4, 6, 8, 10, 12, 14, find the five-number summary and IQR; exclude the median from each half. Add 3 to every value. Predict and verify the new summary, IQR, and range. Explain which measurements stay unchanged.

Grade 6 cumulative check

Tutor choice: use this check in place of the routine questions, or in a separate assessment sitting. Keep the two daily challenges for discussion.

1. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

2. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

3. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

4. Find 140% of 30 points.

5. 12 shells are 30% of a collection. Find the whole collection.

6. How many $\frac{3}{4}$-liter groups fit in $\frac{7}{8}$ liter? Include part of a group.

7. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

8. Which is the reciprocal of $2\ \frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

9. Three notebooks cost $2.45 each. What is the total cost?

10. Find $9.6 \div 8$.

11. Find $1{,}344 \div 32$. Verify by multiplication.

12. Expand $2(x + 7)$.

13. Solve $x + 9 = 23$.

14. Solve $(\frac{3}{4})\ x = 12$.

15. A parallelogram has base 9, height 4, and slanted side 5 cm. Find its area.

16. A triangle has base 9 cm and height 4 cm. Find its area.

17. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

18. Order -7, 2, -1 from least to greatest.

19. Reflect (-3, 4) across the y-axis.

20. Write an inequality for x that means at least 9. Is the boundary included?

21. Find the mean of 2, 3, 7, 8.

22. Find the median of 9, 1, 5, 3, 7.

23. Find Q1, Q3, and IQR for 1, 2, 4, 5, 7, 8, 10, 11. Exclude the median if the count is odd.

24. Find the MAD of 1, 3, 5, 7.

Week 9 reference

Data vary. Center describes a typical value, and spread describes the variation. A display makes both easier to see.

Data displays and center

A statistical question expects different answers. Mean shares the total equally; median locates the ordered middle.

Example: Find the mean of 3, 4, 5, and 8.

The total is 3 + 4 + 5 + 8 = 20.

There are four values, so share 20 among four groups.

Mean = 20 ÷ 4 = 5. The deviations −2, −1, 0, 3 balance to zero.

Result: 5

Spread and box plots

Center describes a typical value. Spread describes how far values are apart or away from the center.

Example: Find the IQR for 2, 4, 6, 8, 10, 12.

Split the ordered data into 2, 4, 6 and 8, 10, 12.

The two half-medians are Q1 = 4 and Q3 = 10.

IQR = 10 − 4 = 6.

Result: 6

Words to use

statistical question: A question answered with data that are expected to vary.

mean: The total shared equally among all data values.

median: The middle of ordered data. With two middles, use their mean.

histogram: A display counting values in number intervals.

range: Largest value minus smallest value.

Data, center, and spread

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 11 × 3

2. 11 × 7

3. 11 × 4

4. 11 × 9

5. Reflect (-3, 4) across the y-axis.

6. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

7. 5 × 2

8. Find the mean of 1, 4, 5, 6.

9. Find the mean of 2, 3, 7, 8.

10. Find the median of 9, 1, 5, 3, 7.

11. Find the median of 8, 2, 4, 6.

12. Four scores have mean 5. Three are 2, 4, 7. Find the fourth.

13. Luis finds the median of 9, 2, 6, 3 by averaging 2 and 6. He gets 4. What should he change?

14. Choose the statistical question: (A) How many pages are in one named book?(B) How many pages are in each of 10 books? Explain.

15. Make a dot plot for 3, 3, 4, 5, 5, 5, 7. State the total dots and most frequent value.

16. Find histogram frequencies for 2, 3, 4, 4, 5, 6, 7, 7 using bins 2 to under 4, 4 to under 6, and 6 to under 8.

17. Find the mean of 4, 6, 8, 10. Explain it as a fair share.

18. Order 10, 4, 7, 6, 7, 11 and find the median.

CHALLENGE / Explain your plan and check the result

C1. Two teams record scores. Team A has three scores with mean 8; Team B has five scores with mean 12. Find the combined mean. Explain why simply averaging 8 and 12 gives the wrong result.

C2. Create four nonnegative whole-number data values with mean 5 and median 5, but range 8. Give one set and verify all three conditions. Then give a different set with the same mean and median but smaller range.

Data, center, and spread

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find $|-12|$ and describe it as a distance.

6. Find $2.1 \div 0.07$.

7. $72 \div 6$

8. Find the MAD of 3, 3, 9, 9.

9. Find the range of 2, 5, 8, 11.

10. Find Q1, Q3, and IQR for 1, 2, 4, 5, 7, 8, 10, 11. Exclude the median if the count is odd.

11. Find the MAD of 1, 3, 5, 7.

12. A: 3, 4, 5. B: 1, 4, 7. Compare their means and ranges.

13. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

14. Choose the statistical question: (A) How many pages are in one named book?(B) How many pages are in each of 11 books? Explain.

15. Make a dot plot for 4, 4, 5, 6, 6, 6, 8. State the total dots and most frequent value.

16. Find histogram frequencies for 3, 4, 5, 5, 6, 7, 8, 8 using bins 3 to under 5, 5 to under 7, and 7 to under 9.

17. Find the mean of 5, 7, 9, 11. Explain it as a fair share.

18. Order 11, 5, 8, 7, 8, 12 and find the median.

CHALLENGE / Explain your plan and check the result

C1. Data are 1, 2, 4, 5, 7, 8, 10, 11. Find the five-number summary and IQR. Replace 11 with 31 and recompute only the range and IQR. Explain why one spread measure changes and the other does not.

C2. Two box plots have summaries A: (0, 2, 5, 8, 10) and B: (0, 4, 5, 6, 10). Compare their medians, ranges, and IQRs. Can you determine which has more observations from these summaries alone? Explain.

Data, center, and spread

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 11.2×10

2. $1100 \div 100$

3. Double 11.5

4. Half of 66

5. Find the range of 0, 2, 5, 10.

6. Find the MAD of 2, 5, 5, 8.

7. Find $|-3|$ and describe it as a distance.

8. A prism has right-triangle ends with sides 3, 4, 5 cm and length 6 cm. Find surface area.

9. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

10. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

11. 7×4

12. A box is 7 cm by 4 cm by 3 cm. Find its volume.

13. Find two sets of five different whole numbers with minimum 2, maximum 12, median 5, and mean 6.

14. Choose the statistical question: (A) How many pages are in one named book?(B) How many pages are in each of 12 books? Explain.

15. Make a dot plot for 5, 5, 6, 7, 7, 7, 9. State the total dots and most frequent value.

16. Find histogram frequencies for 4, 5, 6, 6, 7, 8, 9, 9 using bins 4 to under 6, 6 to under 8, and 8 to under 10.

17. Find the mean of 6, 8, 10, 12. Explain it as a fair share.

18. Order 12, 6, 9, 8, 9, 13 and find the median.

19. Four values have mean 10. Three are 7, 9, 11. Find the missing value.

20. Find the range of 5, 7, 9, 12, 15. Which values determine it?

21. Find the five-number summary and IQR of 5, 6, 8, 10, 12, 14, 16. Exclude the median from each half.

22. Find the five-number summary and IQR of 5, 7, 8, 10, 12, 14, 15, 17.

23. Find the mean and MAD of 5, 7, 11, 13. Show all four distances.

CHALLENGE / Explain your plan and check the result

C1. A histogram uses bins 0 to under 5, 5 to under 10, and 10 to under 15. Data are 0, 4, 5, 5, 9, 10, 12, 14. Find the three frequencies and the fraction in the highest bin. Explain where the value 5 belongs and why.

C2. Six game scores are 2, 4, 6, 6, 8, and 10. A scorekeeper accidentally records 10 as 40. Find the original and incorrect means and medians. Which calculation provides a useful warning that one entry may be wrong?

Data, center, and spread

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 1148 to the nearest hundred.

2. Round 11.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. Find the range of 0, 2, 5, 10.

6. Find the MAD of 2, 5, 5, 8.

7. Data are 1, 2, 4, 4, 5, 6, 6. How many dots belong above 4 in a dot plot?

8. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

9. Order −3, 1, −6 from least to greatest.

10. Find |5| and describe it as a distance.

11. Reflect (2, -5) across the x-axis.

12. Find the distance from (-2, -3) to (-2, 2).

13. Find Q1, Q3, and IQR for 0, 2, 3, 5, 7, 9, 10, 12. Exclude the median if the count is odd.

14. Zoe averages −3, −1, 1, and 3 to find MAD. She gets 0. What should she average?

15. A $40 item is 25% off. Find the sale price.

16. Solve 3(x + 2) = 18.

17. Choose the statistical question: (A) How many pages are in one named book?(B) How many pages are in each of 13 books? Explain.

18. Make a dot plot for 6, 6, 7, 8, 8, 8, 10. State the total dots and most frequent value.

CHALLENGE / Explain your plan and check the result

C1. For 3, 5, 7, 9, 11, find the mean, each distance from the mean, and MAD. A student averages the signed changes − 4, −2, 0, 2, 4 and gets 0. Explain the error and give the correct averaging calculation.

C2. A class records weekly practice hours 2, 2, 4, 6, 6, 8. If every student practices 2 extra hours next week, find the new mean, median, and range. Predict first and justify without recalculating every measure from scratch.

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Find the range of 0, 2, 5, 10.

2. Find the MAD of 2, 5, 5, 8.

3. Data are 1, 2, 4, 4, 5, 6, 6. How many dots belong above 4 in a dot plot?

4. Four scores have mean 7. Three are 4, 5, 9. Find the fourth.

5. A triangle has base 9 cm and height 4 cm. Find its area.

6. Find the distance from (-4, 2) to (3, 2).

7. Write an inequality for x that means no more than 12. Is the boundary included?

8. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

9. How many $\frac{1}{3}$-meter pieces fit in 2 meters?

10. Luis finds the median of 9, 2, 6, 3 by averaging 2 and 6. He gets 4. What should he change?

11. Choose the statistical question: (A) How many pages are in one named book?(B) How many pages are in each of 14 books? Explain.

12. Make a dot plot for 7, 7, 8, 9, 9, 9, 11. State the total dots and most frequent value.

13. Find histogram frequencies for 6, 7, 8, 8, 9, 10, 11, 11 using bins 6 to under 8, 8 to under 10, and 10 to under 12.

14. Find the mean of 8, 10, 12, 14. Explain it as a fair share.

15. Order 14, 8, 11, 10, 11, 15 and find the median.

16. Four values have mean 12. Three are 9, 11, 13. Find the missing value.

17. Find the range of 7, 9, 11, 14, 17. Which values determine it?

18. Find the five-number summary and IQR of 7, 8, 10, 12, 14, 16, 18. Exclude the median from each half.

CHALLENGE / Explain your plan and check the result

C1. Data are 2, 4, 6, 8. Find the mean, median, and MAD. Duplicate every observation so the new data contain eight values. Predict and verify which of the three measures changes.

C2. A data set has five whole-number values with minimum 2, maximum 10, median 6, and mean 6. Give one possible set. Find its range and five-number summary using the exclude-the-median rule. Explain whether your set is the only possible one.

Scale and proportional relationships

20 questions, including 2 challenges. Write your work and answers on separate paper.

R1. A rectangle is 8 cm by 2 cm. Find its area.

R2. List multiples of 4 and 6. Find the least positive common multiple.

R3. Data are 1, 2, 4, 4, 5, 6, 6. How many dots belong above 4 in a dot plot?

R4. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

P1. A 4 cm by 6 cm rectangle is copied with scale factor 2. Find the new side lengths and compare areas.

P2. A triangle has side lengths 6, 8, and 10 cm. Copy it with scale factor $\frac{1}{2}$. Find its side lengths. What happens to its angles?

P3. A map uses 1 cm for 25 km. Two towns are 3.6 cm apart on the map. Find their actual distance.

P4. A drawing has area 5 square cm. A scaled copy has all lengths three times as large. Find its area.

P5. A table pairs x = 1, 2, 4 with y = 2.5, 5, 10. Find k and write y = kx.

P6. Compare two straight-line graphs: A passes through (0, 0), (1, 2), (2, 4). B passes through (0, 3), (1, 5), (2, 7). Which is proportional?

P7. A ride costs $4 to start and $3 per mile. Write cost c for m miles. Is it proportional?

P8. A faucet fills $\frac{3}{4}$ liter in $\frac{1}{2}$ minute at a constant rate. Write liters L after t minutes.

F. A proportional relationship has (x, y) = (4, 14). Complete: k = 14 ÷ __ = __, so y = __x.

D. Eli triples every length of a drawing. He says its area also triples. Explain and repair his claim.

X1. A rectangle 3 cm by 2 cm is scaled by 2. Find new dimensions and area.

X2. A drawing has area 6 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

S. A proportional relationship has __ for every nonzero input and passes through __.

E. A table pairs x = 1, 3, 5 with y = 5, 11, 17. Is y proportional to x? Support your answer with ratios or the value at zero.

CHALLENGE / Explain your plan and check the result

C1. A map scale is 2 cm for 15 km. A route has map legs of 3.6 cm and 2.4 cm. Find the real length of each leg and the full route. A second map doubles every map length. What scale should it show for the same route?

C2. A6 cm by 10 cm photo is enlarged by scale factor 1.5. Find the new dimensions, perimeter, and area. Compare the perimeter factor with the area factor. Explain why multiplying the original area by 1.5 is insufficient.

Circles and first previews

20 questions, including 2 challenges. Write your work and answers on separate paper.

P1. A circle has diameter 8 cm. Find its circumference in terms of π and using 3.14.

P2. A circle has radius 3 cm. Find its diameter and circumference in terms of π.

P3. Find the area of a circle with radius 4 cm. Give an exact answer and an approximation using 3.14.

P4. A circle has diameter 6 cm. Find its area in terms of π and using 3.14.

P5. Circle A has diameter 4 cm and circumference 4π cm. Circle B has diameter 10 cm and circumference 10π cm. Compare C ÷ d.

P6. A collection grows from 80 cards to 100. Find the percent increase. Then find the percent decrease from 100 back to 80.

P7. Start with 80 tokens. Increase by 25%, then decrease the result by 25%. Find the final amount. Explain why it is not 80.

P8. Use steps on a number line for −2 + 5, −3 + (−4), and 2 − 6.

F. A circle has diameter 16 cm. Complete: radius = __; area = π × __² = __π square cm.

D. Maya finds area from a diameter of 14 cm by writing π × 14² = 196π. Explain and repair it.

X1. A circle has diameter 8 cm. Find circumference in terms ofπ and using 3.14.

X2. A circle has radius 3 cm. Find area in terms ofπ and using 3.14.

X3. A circle has diameter 8 cm. Find radius and exact area.

X4. A count rises from 80 to 100. Find the increase and percent increase.

X5. Start with 120 points. Increase by 10%, then decrease the result by 10%. Find the final amount.

X6. On a number line, start at − 4, move right 7, then left 5. Find the final number and total distance.

S. For circle area, I first identify __. A percent change compares the change with __.

E. A circle has diameter 12 cm. Find its area in terms of π and using 3.14. Show the radius you used.

CHALLENGE / Explain your plan and check the result

C1. A circular pool has diameter 10 m. A square deck is 14 m on each side, with the pool cut out of its center. Find the deck’s area in terms ofπ and using 3.14. Then find the pool’s boundary length. Explain why these answers have different units.

C2. Wheel A has radius 20 cm. Wheel B has twice that radius. Find each circumference and area in terms ofπ. If both roll the same distance without slipping, how many turns does B make while A makes 12? Explain.

Final mixed review

Tutor choice: use this check in place of the routine questions, or in a separate assessment sitting. Keep the two daily challenges for discussion.

1. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

2. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

3. A machine makes 24 labels in 6 minutes. Find labels per minute and minutes per label.

4. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

5. Find $9 \div 40$.

6. Expand $2(x + 7)$.

7. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

8. Find the opposite of -6.

9. Find Q1, Q3, and IQR for 0, 2, 3, 5, 7, 9, 10, 12. Exclude the median if the count is odd.

10. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

Week 10 reference

A fixed multiplier connects scaled lengths and proportional quantities. Circle formulas use radius or diameter in different ways. The final weekdays revisit the entire program.

Scale and proportional relationships

A scaled copy multiplies every length by the same factor. Proportional relationships have the form $y = kx$.

Example: A copy of a 3 cm by 5 cm rectangle uses scale factor 2.

The new lengths are 6 cm and 10 cm.

The original area is 15 square cm; the copy area is 60 square cm.

The area factor is $60 \div 15 = 4 = 2^2$.

Result: 6 by 10 cm; area 60 square cm

Circles and first previews

Circumference measures the boundary. Circle area uses the square of the radius. π is exact; 3.14 is an approximation.

Example: Find area when a circle's diameter is 10 m.

The radius is half the diameter: 5 m.

Square the radius: $5^2 = 25$.

Area = 25π square m, or about 78.5 square m using 3.14.

Result: 25π square m; about 78.5 square m

Words to use

scale factor: The multiplier from an original length to a copy length.

scaled copy: A shape with matching angles and one multiplier for every length.

constant of proportionality: The same y per one x in a proportional relationship.

proportional: Related by a fixed multiplier with zero matched to zero.

radius: Distance from the center to the circle.

Scale, proportions, circles, and review

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 12×3

2. 12×7

3. 12×4

4. 12×9

5. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

6. 4 markers cost $2.60. Find dollars per marker.

7. Find $10 - 1.286$.

8. A prism has right-triangle ends with sides 3, 4, 5 cm and length 6 cm. Find surface area.

9. Order -4, 0, -9 from least to greatest.

10. Find the MAD of 1, 3, 5, 7.

11. A 8 by 12 cm rectangle is copied with scale factor $\frac{1}{2}$. Find the new side lengths.

12. A map uses 1 cm for 20 km. A route is 4 cm long. Find the actual distance.

13. A shape has area 20 square cm. All lengths scale by $\frac{1}{2}$. Find the new area.

14. A proportional table contains (6, 9). Find k in $y = kx$.

15. Eli triples every length of a drawing. He says its area also triples. Explain and repair his claim.

16. A rectangle 5 cm by 4 cm is scaled by 2. Find new dimensions and area.

17. A drawing has area 20 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

18. A map uses 1 cm for 5 km. A route is 4.5 cm on the map. Find actual distance.

CHALLENGE / Explain your plan and check the result

C1. A model bridge is 48 cm long at a scale of 1 cm for 2.5 m. Find the real length. A new model uses 1 cm for 4 m. Find the new model length and the scale factor from the first model to the second.

C2. A recipe uses 3 cups of flour for 8 muffins. A baker needs 20 muffins and has 6 cups. Find the required flour and the shortage. If flour costs $0.80 per cup, find the cost of exactly the missing amount.

Scale, proportions, circles, and review

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

6. Mix A has 2 cups juice and 3 cups water. Mix B has 3 cups juice and 5 cups water. Which has more juice for the same water?

7. Find $3\frac{1}{2} \div \frac{3}{4}$.

8. Solve $1.5x = 4.80$.

9. Find $|0|$ and describe it as a distance.

10. Find the median of 8, 2, 4, 6.

11. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

12. A circle has radius 2 cm. Find circumference exactly and using 3.14.

13. A circle has radius 2 cm. Find area exactly and using 3.14.

14. A circle has diameter 4 cm. Find area exactly and using 3.14.

15. Maya finds area from a diameter of 14 cm by writing $\pi \times 14^2 = 196\pi$. Explain and repair it.

16. A rectangle 6 cm by 5 cm is scaled by 2. Find new dimensions and area.

17. A drawing has area 30 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

18. A map uses 1 cm for 6 km. A route is 5.5 cm on the map. Find actual distance.

CHALLENGE / Explain your plan and check the result

C1. A circular garden has radius 7 m. A2 m gate replaces part of the fence. Using$\pi \approx 3.14$, find the fence length and the cost at $5 per meter. Explain why subtracting the gate from the area would be wrong.

C2. A jacket costs $80. Store A reduces the price by 25%, then adds 10% tax. Store B adds 10% tax first, then reduces the taxed total by 25%. Find both final prices. Explain why these two orders agree.

Scale, proportions, circles, and review

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 12.2 × 10

2. 1200 ÷ 100

3. Double 12.5

4. Half of 72

5. Compare $\frac{2}{4}$ and $\frac{2}{5}$.

6. Yellow : purple is 2 : 5. There are 35 beads. Find both counts.

7. Find 140% of 30 points.

8. Which is the reciprocal of $2\frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

9. Find 2.45 + 4.078.

10. Solve 1.5x = 4.80.

11. Find the surface area of a closed 1 by 4 by 5 cm box.

12. Write an inequality for x that means less than 5. Is the boundary included?

13. A: 3, 4, 5. B: 1, 4, 7. Compare their means and ranges.

14. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

15. A garden is 3 cm by 4 cm on a map where 1 cm means 10 m. Redraw it where 1 cm means 20 m. Find the new drawing dimensions and the actual garden area.

16. A rectangle 7 cm by 2 cm is scaled by 2. Find new dimensions and area.

17. A drawing has area 14 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

18. A map uses 1 cm for 7 km. A route is 2.5 cm on the map. Find actual distance.

19. A proportional table includes(x, y) = (2, 14). Find k, and find y when x = 5.

20. A rental costs $7 to start plus $2 per hour. Write cost c for h hours. Is the relationship proportional?

21. A pump fills 7 liters in 2 minutes. At a constant rate, write L after t minutes and find the amount after 4 minutes.

22. A circle has diameter 14 cm. Find circumference in terms ofπ and using 3.14.

23. A circle has radius 2 cm. Find area in terms ofπ and using 3.14.

CHALLENGE / Explain your plan and check the result

C1. A circular running track has diameter 40 m. Using 3.14, find the distance of 3 full laps plus half a lap. A runner covers it in 7 minutes. Find the average speed in meters per minute.

C2. Four orders cost $18, $22, $26, and $34. Find the mean. A fifth order is added so the new mean is $27. Find its cost and the new median. Explain why the new mean and median differ.

Scale, proportions, circles, and review

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 1248 to the nearest hundred.

2. Round 12.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

6. Find 140% of 30 points.

7. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

8. Find 9.6 ÷ 8.

9. Solve x + 14 = 31.

10. A rectangle is 7 cm by 3 cm. Find its area.

11. Find the mean of 1, 4, 5, 6.

12. Find the median of 10, 3, 5, 7, 2, 9.

13. Find the range of 0, 2, 5, 10.

14. Find the MAD of 2, 5, 5, 8.

15. A map uses 1 cm for 50 km. A route is 2 cm long. Find the actual distance.

16. Maya finds area from a diameter of 14 cm by writing $\pi \times 14^2 = 196\pi$. Explain and repair it.

17. Compute (−3) × 4 and (−12) ÷ (−3).

18. A price rises from $20 to $25. Find the percent increase.

CHALLENGE / Explain your plan and check the result

C1. An elevator starts on floor − 3, travels up 8 floors, then down 6. Find its final floor and total floors traveled. Compare the final position with its distance from zero; explain why neither equals total travel.

C2. A drawing shows a triangle with base 8 cm and height 5 cm. A copy has base 12 cm and is scaled uniformly. Find the copy’s height and area. A student multiplies 20 square cm by 1.5 and gets 30. Explain the missing step.

Weekly check

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Find $\frac{3}{4} \times \frac{4}{5}$.

2. Plot (1, 2), (2, 4), and (4, 8). Do they lie on a line through the origin?

3. Find 25% of 32 stickers.

4. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

5. Find 0.25×0.04.

6. Expand $3(x + 6)$.

7. Find the surface area of a closed 1 by 4 by 5 cm box.

8. Find the opposite of -1.

9. Find the MAD of 3, 3, 9, 9.

10. Eli triples every length of a drawing. He says its area also triples. Explain and repair his claim.

11. A rectangle 9 cm by 4 cm is scaled by 2. Find new dimensions and area.

12. A drawing has area 36 square cm. Every length is scaled by $\frac{1}{2}$. Find the new area and explain the factor.

13. A map uses 1 cm for 9 km. A route is 4.5 cm on the map. Find actual distance.

14. A proportional table includes$(x, y) = (4, 36)$. Find k, and find y when $x = 7$.

15. A rental costs \$9 to start plus \$4 per hour. Write cost c for h hours. Is the relationship proportional?

16. A pump fills 9 liters in 4 minutes. At a constant rate, write L after t minutes and find the amount after 8 minutes.

17. A circle has diameter 18 cm. Find circumference in terms ofπ and using 3.14.

18. A circle has radius 4 cm. Find area in terms ofπ and using 3.14.

CHALLENGE / Explain your plan and check the result

C1. A rectangular plaza is 12 m by 8 m. A circular fountain of diameter 4 m occupies part of it. Using 3.14, find the remaining paving area. Paving costs \$10 per square meter, and a10% discount applies to the paving bill. Find the discounted cost.

C2. A print shop charges Plan A: \$6 setup plus \$0.20 per page, and Plan B: \$0.35 per page. Find both prices for 20 and 60 pages. Find the page count where they agree. Decide whether either plan is proportional to page count and justify.